

A MODEL OF OPTIMAL SCIENCE FUNDING: TARGETING THE CAPABILITY-RESOURCE GAP

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ABSTRACT. The grant funder’s challenge is to maximize the impact of a fixed budget under uncertainty about the researchers who might receive it. We develop a Bayesian model of this challenge: researchers differ in their capabilities and in their resources; research output requires both, and funding compounds capability across rounds. We show that the output-maximizing allocation of a round’s budget funds the capability-resource gap: grants increase with capability and decrease with existing resources. Track records do not pin down the gap, because research output confounds the two; what a funder needs is a signal of capability less confounded by resources than the track record, and peer review of a proposal can provide one. We find that the value of that signal increases with how unequally the gaps are distributed, which capability inequality mainly drives, and with the signal’s informativeness, and that under heavy-tailed capability distributions even a moderately informative signal captures most of that value. Uniform seed grants and lotteries, which forgo targeting, lose little research output exactly where review is worth little, and where informative review makes a partial lottery costly, most of what the lottery loses comes from discarding the review ranking rather than from chance. Two features of a field, the funder’s budget relative to researchers’ resources and the inequality of the gaps, locate a funder on a map of these results.

1. INTRODUCTION

Science funding is large and consequential, and how best to allocate it is still debated. The United States spent close to a trillion dollars on research and development in 2024, roughly a fifth of it funded by the federal government (National Center for Science and Engineering Statistics 2026; National Science Board 2026). Whether that money should be concentrated among a few researchers or spread across many is a long-running dispute (Fortin and Currie 2013; Mongeon et al. 2016; Aagaard et al. 2020; Lauer and Roychowdhury 2021). How well peer review predicts research outcomes is contested: among funded NIH grants, percentile scores predicted subsequent productivity barely better than chance (Fang et al.

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2016), while studies with other designs find modest but real predictive power (Gallo et al. 2014; Li and Agha 2015; Park et al. 2015). Whether review should be replaced, in part, by lotteries has moved from proposal to practice, in forms that range from a tie-break among top-ranked proposals to a draw among all applicants above a threshold (Roumbanis 2019; Heyard et al. 2022; Feliciani et al. 2024). A randomized trial of winning a grant found no clear effect on the recipients’ publications over the following four years (Barnett et al. 2024), and a quasi-experimental study of NIH grants found a small one (Jacob and Lefgren 2011), which raises the more basic question of when an additional grant adds value at all.

These disputes concern a single underlying problem: what a funder with a fixed budget and imperfect knowledge of researchers should do. The empirical evidence cannot settle them on its own, because each study measures one input to that decision, how well scores predict outcomes among funded grants or how research output responds to grant size, and none can say what the funder should do with what it knows. That question depends on what the funder is uncertain about, on what its information is worth, and on how the answer changes with the budget. It is a question for decision theory. This paper treats the funder’s problem as a sequential decision under uncertainty and analyzes it using a Bayesian approach. It answers three questions: whom to fund, what information about researchers is worth, and how seed grants and lotteries perform relative to targeted funding; a fourth, whether to spend early or wait and learn, is treated in Appendix E. The model is deliberately simple, and its contribution is to our theoretical understanding: it derives the output-maximizing allocation, shows how the three disputes are structured, and identifies the two features of a field on which the answer to each of them turns. The objective throughout is aggregate expected research output; diversity of directions, risk, and fairness are outside it, and “optimal” means output-maximizing under that objective.

Consider the challenge of the grant funder, whether a federal agency disbursing billions or a private foundation disbursing millions: you have a fixed budget and must choose how to allocate it across researchers and research projects to create the greatest positive impact. Two intuitive schemes might be considered. Fund the track record: give to those who have produced the most, since past productivity is evidence of future productivity. Fund the under-resourced: give to those who have the least, since a dollar goes further where dollars are scarce. Both, we will show, are suboptimal, and for the same reason: each accounts for only one of the two quantities the impact of a grant depends on.

We model this decision situation as follows. A funder with a fixed budget allocates grants, over multiple rounds, to researchers who differ in two ways: epistemic capability, what a researcher knows and can do, and resources, what a researcher has to work with. Research output requires both and is limited by the scarcer of the two; the funder can observe research output, but not capability or resources themselves. A grant adds to a researcher’s resources in the round it is given and, because doing research builds expertise, raises their capability in later rounds; we call this compounding. Each round, the funder observes the research output that results, updates its beliefs about the researchers, and allocates again.

Our results are as follows. When the funder knows every researcher’s capability and resources, the allocation of a round’s budget that maximizes expected research output funds the capability-resource gap: each funded researcher receives a grant increasing in their capability and decreasing in their existing resources, and the budget determines how many researchers receive funding at all. Each intuitive scheme fails because it accounts for only one of the two quantities, and each can perform worse than simply dividing the budget uniformly across all researchers. In reality the funder lacks direct access to the gap, and track records alone do not pin it down: we show that a funder relying on the track record performs far worse than one that estimates the gap.

Track records do not pin down the gap because research output confounds capability with resources; what the funder needs is a signal of capability less confounded by resources than the track record, and peer review of a proposal can provide one. The value of that signal to the funder is the expected research output it adds by reducing the funder’s uncertainty about the gap, and that value is greatest where capability is unequally distributed and review informative; in fields with heavy-tailed distributions of capability, even a moderately noisy signal from peer review captures most of this value. The same reasoning says how uniform seed grants and lotteries perform relative to targeted funding: each gives up the value of the targeting it replaces, which is small where review is worth little and large where review is worth most. We vary the budget throughout, so our results apply to funders of every size, and the budget is one of the two features that decide every result: targeting matters most per dollar for a small foundation, whose grants are small relative to researchers’ total resources, and whether optimal funding should be concentrated among a few researchers or spread across many depends on the budget relative to those resources. Within the model, no funding mechanism is best in every field; the funder’s budget and the field’s distribution of capability decide which mechanism adds research output. Finally, the two features can be

estimated from what funders already know, and we place representative funding programs on the budget axis of the map they define.

Two assumptions limit these results. First, grants are used up in the round they are given; whatever lasts from a grant (skills, publications, standing) is in the form of naturally compounding researcher capability generated by the chance to produce more research. Second, capability and resources are complements throughout: neither fully substitutes for the other. We defend each assumption where it is used.

We proceed as follows. §2 situates our model in the literatures on funding allocation, peer review, and lotteries. §3 presents our model. §4 derives the optimal allocation. §5 shows what track records can and cannot reveal about the gap, then measures the value of peer review and relates it to the empirical evidence on review. §6 measures how seed grants and lotteries perform relative to targeted funding and asks how widely funds should be spread. §7 discusses implications and limitations, places representative funders on the map the two features define, and states the model’s testable predictions; §8 concludes. Appendix E examines the timing of funding, including the strongest argument for spending early.

2. RELATED LITERATURE

Our contribution sits at the intersection of four literatures: the empirical debate over concentrating or dispersing funds, the evidence on how well peer review predicts research outcomes, the case for funding lotteries, and formal models of science funding.

Whether funders should concentrate resources on a few researchers or spread them across many is a long-running dispute, and concentration is the current practice: the top 1% of NIH principal investigators receive close to 10% of research project grant funding, a share that has risen since 1985 (Katz and Matter 2020; Lauer and Roychowdhury 2021). Empirical studies of funded portfolios generally find diminishing marginal returns to funding per researcher (Fortin and Currie 2013), and reviews of this literature conclude that the evidence favors more spreading than current practice (Mongeon et al. 2016; Aagaard et al. 2020). Concentration is also self-reinforcing: early grant winners accumulate advantages beyond the grant itself (Merton 1968; Bol et al. 2018). The effect of a grant on its recipient’s research output is itself modest in the studies that can measure it: a quasi-experimental study of NIH grants found about one additional publication per grant (Jacob and Lefgren 2011), and a randomized trial found no clear effect over four years (Barnett et al. 2024). This dispute typically concerns the returns to funding: whether research output per researcher increases or decreases as grants concentrate. In our

model, the answer also depends on the size of the funder’s budget relative to researchers’ total resources: with a budget large enough to relieve most bottlenecks, the optimal allocation itself reaches nearly every researcher, so little is lost by spreading funds widely; with a tight budget, effectiveness requires concentrating funds on the researchers with the largest capability-resource gaps (§6). The value of any targeting, in turn, depends on what the funder can know about researchers (§5).

How well peer review predicts research outcomes is contested. Among funded NIH grants, percentile scores predict subsequent productivity barely better than chance (Fang et al. 2016), and reviewers assessing the same application agree only weakly (Pier et al. 2018). Studies with other designs find real but modest predictive power: better scores predict more publications, citations, and patents, including under controls for investigator history and institution (Li and Agha 2015), and a natural experiment in funding finds that higher-scored proposals produce more than lower-scored ones (Park et al. 2015); the null findings among funded grants are also what restriction of range predicts, since nearly all funded applications score in the top third (Lindner and Nakamura 2015). Both sides of this dispute measure review’s accuracy: how well scores predict outcomes among funded applicants. Our model instead measures review’s value: the expected research output that review adds by reducing the funder’s uncertainty about the capability-resource gap. That value depends on the field; the same noisy signal can be worth a great deal in one field and nothing in another (§5), and no single number settles the question.

Doubts about review have made lotteries a live proposal. Advocates argue that randomizing among applications that pass a review threshold saves review costs (Herbert et al. 2013), reduces bias (Ginther et al. 2011; Wittman et al. 2019; Boudreau et al. 2016), and honestly reflects review’s limited ability to discriminate among proposals that pass a review threshold (Pier et al. 2018; Fang and Casadevall 2016; Avin 2019a; Roumbanis 2019; Shaw 2023); a parallel case on the same grounds has been made for abolishing prepublication peer review of articles (Heesen and Bright 2021). Several funders now run partial lotteries (Liu et al. 2020; Heyard et al. 2022; Luebber et al. 2025). The existing formal argument for lotteries measures what a lottery saves: Gross and Bergstrom (2019) model grant competition as a contest in which scientists invest costly effort in proposals, show that when only a small fraction of proposals is funded this effort can rival the scientific value of the research funded, and show that a partial lottery among proposals above a threshold removes the link between the wasted effort and the fraction funded. What has not been measured is what a lottery gives up: the value of targeting. Our model measures that value. It is negligible where

budgets are ample, review is uninformative, or capability is evenly spread, and substantial where capability is heavy-tailed and budgets are tight (§6). Together the two measures let a funder weigh the choice field by field.

Formal models of science funding have concentrated on the incentive costs of competition and on the value of exploration (Weisberg and Muldoon 2009). Sakamoto (2023) compares review with baseline funding against review with a lottery and finds baseline funding better in many situations; our seed grants play the role of baseline funding, and the comparison here turns on the field. Beyond the contest model above, Gross and Bergstrom (2025) find, in a game-theoretic model of contests for scarce rewards, that intensifying competition pushes scientists toward higher-risk, higher-return projects. Avin (2019b) simulates funding strategies on an epistemic landscape and finds that allocation with an explicitly random component significantly outperforms a peer-review-like strategy on large landscapes, because review concentrates effort on regions already known to be productive. Sorzano and Pueche-Granados (2026) model allocation as a decision under heavy-tailed uncertainty about project value, and combine a citation-based signal with a lottery weighted by that signal; they find more room for lotteries than we do, and the difference is instructive: their heavy tail is in observed impact, whereas ours decomposes research output into unobserved capability and resources, and that decomposition is what gives review its value in our model. In each of these models, funding raises a single quantity, a project’s chance of success or a researcher’s productivity, and none treats the timing of spending as the funder’s choice. In our model, a researcher’s research output increases in both capability and resources but is limited by the scarcer of the two, and the funder observes neither directly. This formulation lets us address questions raised by all four literatures: how widely funds should be spread (§6), what peer review is worth (§5), and when lotteries lose little (§6).

3. OUR MODEL

A funder faces a population of n researchers over T funding rounds. Researcher i has capability K_i and resources R_i . Initial capabilities and resources are drawn from Pareto distributions with tail parameters α_K and α_R ; smaller tail parameters produce more unequal fields (below 2 the variance is infinite), and the two are coupled through a Gaussian copula with parameter ρ , so that capability and resources can be associated across researchers (a Pearson correlation between Pareto variables is undefined when their variances are infinite; $\rho = 0.5$ and 0.8 correspond to rank correlations of 0.48 and 0.79). The funder knows these distributions; it never observes any researcher’s capability or resources directly. Capability is a state,

not a trait: it holds whatever a researcher has accumulated that raises research output, expertise, methods, a trained group, and doing research adds to it.

In each round, researcher i produces research output¹ at random, in discrete units, with expected research output²

$$\lambda_i = A \Lambda(K_i, R_i),$$

where A is a productivity constant and Λ is a production function that increases in both arguments but is limited by the scarcer of the two. Our main analysis takes Λ to be proportional to the harmonic mean of the two inputs:

$$\lambda_i = \frac{AK_i R_i}{K_i + R_i}.$$

The harmonic mean makes capability and resources complements without making either an absolute requirement: a capable researcher with few resources still produces, by working more slowly or borrowing equipment, but no surplus of one input fully substitutes for a shortage of the other. Appendix C repeats our analysis for a family of production functions running from Cobb-Douglas, under which either input can substitute for the other, to Leontief, under which neither can.

Capabilities increase through research. Between rounds,

$$K_i \leftarrow K_i + \epsilon \Lambda(K_i, R_i),$$

where ϵ is the compounding rate: doing research builds expertise, and the same bottleneck that limits research output limits growth. Growth requires both inputs, and capability never declines. Resources, by contrast, do not accumulate: resources in a round are baseline plus grant, $R_i = R_{i0} + g_i$. The baseline R_{i0} recurs every round, representing the time and support a researcher has independent of the funder, while grants are consumed in the round they are given. Whatever lasts from a grant (skills, publications, standing) enters the model as capability.

The funder's total budget is fixed before any outcomes are observed. We parameterize it by a budget scale b : at $b = 1$, the budget equals the community's expected baseline resources for one round, and the budget scales linearly in b (Appendix F). The budget does not grow with the horizon: a longer horizon varies the timing of spending, never its amount. Strategies that do not plan ahead spend an equal share of the budget each round. Forward-looking strategies may time their spending freely: each round, they re-plan the entire remaining budget over the remaining horizon and execute the current round's allocation.

¹ We are agnostic about the interpretation of research output units: they may be interpreted as publications, as findings, or as units of scientific value.

² Research output in each round is drawn from a Poisson distribution with rate λ_i , independently across researchers and rounds given the state.

The funder learns as it goes. It observes each researcher’s track record, the research output they have produced to date, as it accumulates, and two further signals are available. A resource signal, observed once at the start, gives a noisy estimate of each researcher’s baseline resources (noise τ_R), as one might obtain from institutional information. A review signal, also observed once at the start by the strategies that use it, gives a noisy estimate of capability (noise τ_K), as one might obtain from proposal evaluation and expert judgment. Formally, the resource signal is $R_{i0} + \eta_i$ with $\eta_i \sim N(0, \tau_R^2)$, and the review signal is $K_i + \xi_i$ with $\xi_i \sim N(0, \tau_K^2)$, independent across researchers; both are held fixed over the horizon, so a reviewer’s misjudgment of a researcher persists across rounds, and only the research output record accumulates (Appendix D reports the variant in which review is redrawn every round). The smaller τ_K , the more informative the review. Bayesian strategies hold a posterior over each researcher’s capability and baseline resources, computed from these likelihoods and the Poisson research output likelihood by importance sampling with M particles drawn from the prior ($M = 1,000$ for Figures 4, 5, and 10 and for Appendix D, and 200 elsewhere; Appendix F checks convergence in M), update it after each round, and allocate each dollar to the researcher whose expected research output, given the posterior, rises most with it. The forward-looking strategies are certainty-equivalent receding-horizon planners: each round they choose the current allocation together with a plan for the remaining budget so as to maximize expected research output over the remaining horizon, treating future research output as equal to its expectation, and they re-plan after observing the round’s research output. They do not value the information their own grants will generate (Appendix E). The complete-information benchmark applies the gap rule of §4 to each round’s equal share of the budget with the true capabilities and resources. The unit of analysis throughout is the researcher, not a researcher-project pair: this is a model of investigator-level funding.

We compare the strategies in Table 1. The Bayesian strategies differ along two dimensions: information, with or without the review signal, and planning, myopic (optimizing the current round’s expected research output) or forward-looking (optimizing expected research output over the whole remaining horizon). Three baselines complete the set: no funding, against which every result is normalized; uniform funding, which splits each round’s budget equally; and track-record funding, which allocates in proportion to the previous round’s research output. Seed grants and lotteries, which modify these strategies, are introduced in §6.

Each of our conclusions rests on a proof or on simulations across the following parameter ranges; default values and the settings behind each figure are in

Strategy	Information	Planning
No funding	none	none
Uniform	none	none
Track record	track record	none
Myopic, without review	track record + resource signal	current round
Forward, without review	track record + resource signal	remaining horizon
Myopic, with review	track record + resource signal + review	current round
Forward, with review	track record + resource signal + review	remaining horizon

TABLE 1. The funding strategies compared.

Appendix F. The tail parameters α_K and α_R range from 1.3 to 3.5. Empirical distributions of research productivity have Pareto tail parameters near 1 to 2 (Lotka 1926; Nicholls 1989); since productivity is an imperfect guide to capability (§5), our range extends well beyond those estimates toward more equal fields. The budget scale b ranges from 0.01 to 3, from a funder whose budget is small beside the community’s own resources to one able to relieve most researchers’ bottlenecks; §5 and Appendix E also examine larger budgets. The compounding rate ϵ ranges from near zero, where funding raises only current research output, to 0.85, where this round’s funding substantially raises later capability. The review noise τ_K ranges from 0.05, nearly perfect review, to 20, nearly uninformative review, a range wide enough to contain the noise implied by empirical estimates of review’s predictive accuracy (§5). The copula parameter ρ ranges from -0.5 to 0.8 , and the horizon T from one round to ten.

Randomness enters through the initial population draws, the signals, and realized research output; capability growth follows expected research output, so a researcher’s accumulation depends on their capability and resources rather than on luck in a given round. We score each strategy by expected research output and compare strategies on identical simulated populations, and we report each strategy’s improvement over no funding in percent (Appendix F). Numbers quoted in the text and footnotes are read from the figure named at the stated setting and illustrate a comparison; the claims are the qualitative comparisons, which hold across the ranges of our parameter sweep (Appendix D).

With the model in place, the funder’s problem is well defined: choose grants, given beliefs, to maximize expected total research output over the horizon. Next we solve the case in which the funder has complete information regarding researchers’ capabilities and resources.

4. THE OPTIMAL ALLOCATION: FUND THE CAPABILITY-RESOURCE GAP

First, consider the simple case where the funder has complete information regarding researchers’ capabilities and resources. Further, we initially restrict our

attention to a single round of funding, setting aside for now the growth of capability across rounds. The funder's problem is to divide the round's budget, B , among researchers so as to maximize expected research output:

$$\max_{g_1, \dots, g_n \geq 0} \sum_i A \Lambda(K_i, R_{i0} + g_i) \quad \text{subject to} \quad \sum_i g_i \leq B.$$

Now, consider the marginal value of a dollar for such a funder. Under the harmonic production function,

$$\frac{\partial \lambda_i}{\partial g_i} = \frac{A K_i^2}{(K_i + R_{i0} + g_i)^2}.$$

The marginal value of funding researcher i increases with their capability and decreases with the resources they already hold, including grants. An allocation is optimal exactly when no dollar can be moved to a researcher who can produce more value with it, that is, when marginal values are equal across all funded researchers and no unfunded researcher's marginal value exceeds that common value.

Proposition 1 (the gap rule). *For every researcher, the optimal grant fills the gap, if any, between the resources they hold and a target proportional to their capability:*

$$g_i^* = \max(c K_i - R_{i0}, 0),$$

where the constant $c > 0$ is the value at which the grants sum to the budget.³

Each researcher's target is c times their capability, and their gap is the difference between that target and the resources they hold. The rule grants every researcher with a positive gap exactly their gap and grants nothing to the rest. When the budget is small, c is small: targets are low and few researchers have positive gaps. When the budget is large, c is large and funding extends across nearly the whole population (Corollary 1, Appendix A). In both cases the funded researchers are those with the least resources relative to their capability, and these are the researchers for whom a dollar has the greatest marginal value.⁴ The condition

³ $c = \sqrt{A/\nu} - 1$, with ν the common marginal value at the optimum. The proof is in Appendix A, and it does not depend on the harmonic form: for any production function (the rule that maps capability and resources to expected research output) with constant returns to scale and diminishing returns to resources, $F(K, R) = A K h(R/K)$ with h increasing and strictly concave, equal marginal value among the funded means a common ratio of resources to capability, and the same rule follows with $c = (h')^{-1}(\nu/A)$. The harmonic mean is the instance we simulate.

⁴ The optimality condition behind the rule, funding to a common marginal value, is the water-filling solution familiar from information theory (Cover and Thomas 2006). If all researchers shared one capability, the rule would reduce to filling each researcher's gap below one common target, the poverty-gap transfer of development economics (Foster et al. 1984); our rule scales the target with capability.

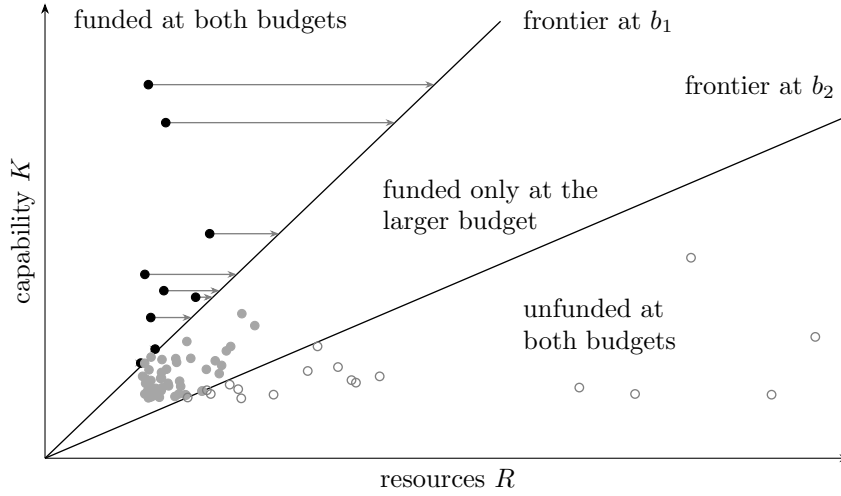


FIGURE 1. The gap rule in one population at two budgets. Each point is a researcher, placed by resources and capability. The two lines are the funding frontiers $R = cK$ at a smaller budget b_1 and a larger budget b_2 ; a larger budget raises c and flattens the frontier. Researchers to the left of a frontier are funded at that budget: black points at both budgets, gray points only at the larger budget, open points at neither. Each arrow is a grant at the smaller budget, which moves its researcher exactly to that budget's frontier; at the larger budget every black and gray researcher receives a grant that moves it to the outer frontier (arrows not drawn).

$R_{i0} < cK_i$ draws a frontier through the population, the funding frontier, separating the funded from the unfunded; a larger budget raises c and moves the frontier outward, so that more researchers are funded and each funded researcher receives more (Figure 1).

The gap rule explains why each of the intuitive funding schemes discussed earlier fails. Funding the track record fails because research output reflects resources as well as capability: a productive researcher whose resources approach or exceed their target has a small or negative gap, and an additional dollar adds little to their research output. The failure is not merely a shortfall from the optimum: because track-record funding channels money toward researchers whose bottleneck is not resources, it can produce less research output than spreading the budget evenly, that is, than not targeting at all. Funding the under-resourced fails for the opposite reason: scarcity is no evidence of capability, and dollars given to researchers of low capability add little research output however scarce their resources. It, too, can produce less than not targeting at all.⁵ (Figure 11, Appendix A, shows both schemes on the population of Figure 1.) Each scheme accounts for one of capability

⁵ Examples 1 and 2 in Appendix A make both failures exact with two researchers.

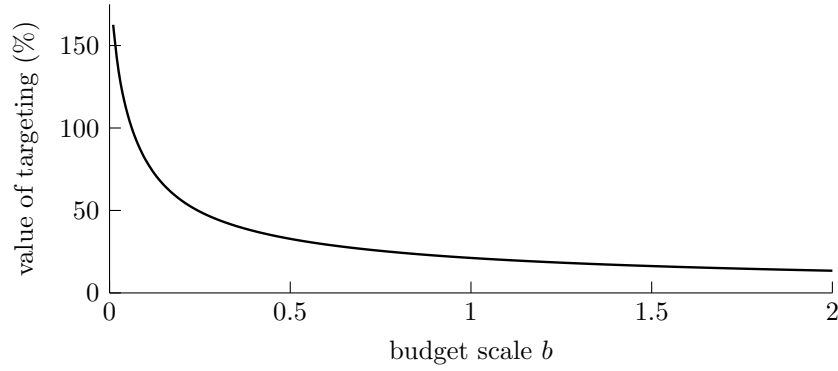


FIGURE 2. The value of targeting decreases in the budget. The additional expected research output of the optimal allocation over uniform funding, as a fraction of uniform funding’s gain over no funding, against the budget scale b , for one population drawn from the default distributions. Corollary 2 proves that the value vanishes as the budget grows; the decrease throughout is a numerical finding (Appendix A).

and resources, whereas the marginal value of a dollar is determined by how a researcher’s resources compare with their capability, and so responds to both.

How much targeting matters depends on the budget. A researcher’s expected research output increases with every dollar of resources, but with diminishing marginal returns, and it never exceeds AK_i : expected research output is bounded above by a quantity set by capability, which it approaches but does not attain. When the budget is large, optimal and uniform funding therefore differ little in the research output they produce: under either, every researcher’s research output is near its upper bound, and the difference between the two vanishes as the budget increases whenever capability bounds research output, as it does under the harmonic mean (Corollary 2, Appendix A); that it also decreases throughout, measured against uniform funding’s own gain, is a numerical finding that held in every population we sampled. When the budget is small, dollars are scarce, researchers differ in the marginal value of a dollar, and research output depends on where each dollar goes. Targeting matters where the budget is small relative to researchers’ resources and little where the budget is large (Figure 2). This dependence on the budget recurs later: it sets how much research output is lost when seed grants and lotteries override the optimal allocation, and it determines when optimal funding concentrates rather than spreads (both §6).

Proposition 1 concerns one round. Over several rounds a grant also raises the recipient’s future capability, and the allocation that maximizes research output over the whole horizon need not take the gap form. Numerically it nearly does.

Over two rounds with complete information, the two-round optimum, free to choose its recipients and how much of the budget to spend in each round, exceeds the gap rule applied round by round with an even split by about one percent of the research output that funding adds or less, at every capability tail, compounding rate, and budget in our parameter sweep, and the two allocations’ first-round grants are nearly identical (Table 2, Appendix A). We therefore use the round-by-round gap rule as the complete-information benchmark throughout, and “the optimal allocation” below refers to it.

To use the gap rule, however, a funder requires an estimate of the gap between researchers’ capabilities and their resources. Next we turn to how such estimation can work: first from the track record alone, then with peer review added.

5. WHAT THE FUNDER CAN LEARN: TRACK RECORDS AND PEER REVIEW

What the track record can and cannot supply. The funder observes track records: each researcher’s realized research output, round by round. A track record is evidence about the capability-resource gap, and the question is how far that evidence goes.

Consider first what research output can identify in principle. Expected research output is one number, produced by capability and resources jointly: every pair of capability and resources with $AKR/(K + R) = \lambda$ produces the same expected research output λ . A researcher of high capability and modest resources and a researcher of modest capability and abundant resources can therefore produce identical records. Research output does carry some information about capability, a lower bound: since expected research output never exceeds AK , a researcher whose expected research output is λ has capability above λ/A , and realized research output, a Poisson draw around λ , bounds it less. Beyond that bound, the record cannot separate the two researchers above, and these are exactly the researchers the gap rule treats most differently: the first may have the largest gap in the population, the second may have none. Research output, however carefully recorded, does not pin down what the funder most needs to know: the gap.

Our model measures how much research output this loses. A Bayesian funder without review, which holds the track record and the noisy resource signal but no review score (§3), allocates far from the gap rule, and further rounds of records improve its allocation only slowly; its research output falls well short of what a funder with complete information would produce.⁶ This shortfall is the most that

⁶ At the default parameters, the correlation between its grants and the optimal grants is 0.08 in the first round and reaches only 0.32 by round twenty; its shortfall below the complete-information optimum, as a share of what that optimum adds, falls from 46 percent in round one to 19 percent in round twenty, against 8 and 3 percent for the funder with a single review score

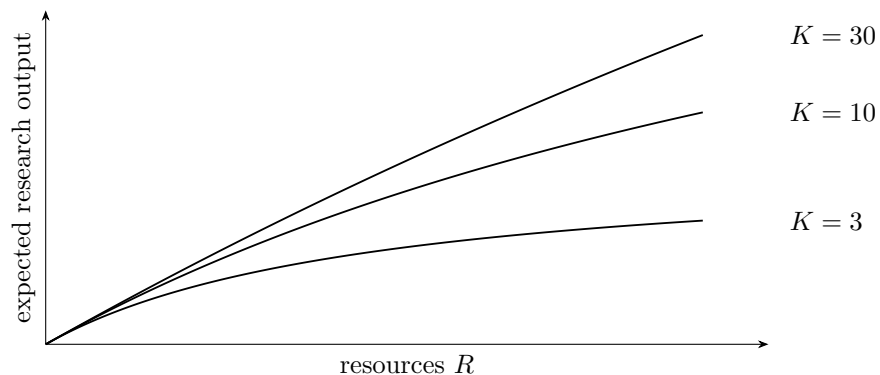


FIGURE 3. On small grants, research output carries little information about capability. Expected research output against resources for researchers of capability $K = 3, 10$, and 30 . Where resources are small relative to capability the curves coincide, so researchers of very different capability produce nearly identical research output; the curves separate only where resources are comparable to capability.

any further information could recover; we ask next how much of it a review signal recovers.

The record is least informative exactly where the gap is largest. When a researcher's resources are small relative to their capability, expected research output is approximately AR : proportional to resources, nearly independent of capability (Figure 3). On a small grant, a researcher of capability 1 and a researcher of capability 100 produce nearly indistinguishable research output. On small grants, therefore, research output carries little information about capability; separating capability from resources by observing research output requires large grants, comparable to capability itself. Appendix E measures this and its consequences for the timing of spending.

What the record cannot supply, then, is a signal of capability less confounded by resources than the record itself. Peer review of a proposal can provide such a signal under the right conditions.

Peer review: what a capability signal is worth. Peer review enters our model as a noisy signal of each researcher's capability, observed once at the start (§3); a score that partly reflects resources is a robustness check in Appendix D. The value of review is the additional expected research output of a funder allocating

observed at the start (Figure 4). Over two rounds, its research output falls short of the complete-information research output by 26 percent of the research output that complete-information funding adds (44 percent under heavy-tailed capability with the same mean).

with the review signal over the same funder allocating without it, the value of the information the score carries (Howard 1966).

The direction of review’s effect is what one would expect: a more informative signal brings the funder’s allocation closer to the optimal allocation and its research output closer to what complete information would deliver, and with a reliable enough signal the funder recovers most of the research output that a funder without review fails to obtain.⁷ Two further effects are less obvious: (1) a funder without review does not catch up, and (2) what a review signal is worth varies widely with the field. The first effect has a structural cause. Capabilities compound across rounds while resources do not accumulate, so each researcher’s research output approaches the level their resources sustain, and research output at that level carries less and less information about capability. Further rounds of records do improve the funder’s allocation, but slowly: after twenty rounds of records the funder’s allocation remains further from the optimal allocation than a funder with the review signal is after one (Figure 4).

Our main result is how much review is worth in different fields (Figure 5). Review is worth most where capability is heavy-tailed and the budget is tight, and in the heavy-tailed case even a signal of moderate informativeness captures most of its value. The reason is twofold. Where capability is heavy-tailed, most of the research output a funder can add comes from getting resources to the few researchers of unusually high capability. And precisely those researchers are the easiest to identify: they stand far apart from everyone else, so even a noisy signal separates them. Identifying the most capable researchers captures most of the value and requires the least information. Making review more reliable beyond a modest level therefore adds little.⁸ In the other fields, review is worth less. Where capability is spread more evenly, no researcher matters much more than another, so review of any informativeness adds little.⁹ And where the budget is ample, optimal and uniform funding produce nearly the same research output, so information about

⁷ At the default parameters, the correlation between the funder’s grants and the optimal grants increases from 0.21 at the noisiest signal in our range to 0.97 at the most reliable; under heavy-tailed capability ($\alpha_K = 1.3$), from 0.31 to 0.99. The research output shortfall relative to the complete-information research output decreases from 25 to 2.3 percent of the research output that complete-information funding adds; under heavy-tailed capability, from 37 to 1.3 percent.

⁸ At the default signal noise, review retains 82 percent of its maximum value under heavy-tailed capability ($\alpha_K = 1.3$), 72 percent at the intermediate distribution ($\alpha_K = 2$), and 45 percent where capability is spread more evenly ($\alpha_K = 3.5$); at three times the default noise, 52, 33, and 8 percent. The three distributions have the same mean capability, so the tail parameter varies how dispersed and how heavy-tailed capability is, not its level; Appendix D reports the sweep at a fixed Pareto scale, which confounds inequality with mean capability and overstates the heavy-tailed field’s advantage.

⁹ At $\alpha_K = 3.5$, review’s value at its maximum is 8.5 percent of the research output that funding with review adds, against 44 percent under heavy-tailed capability.

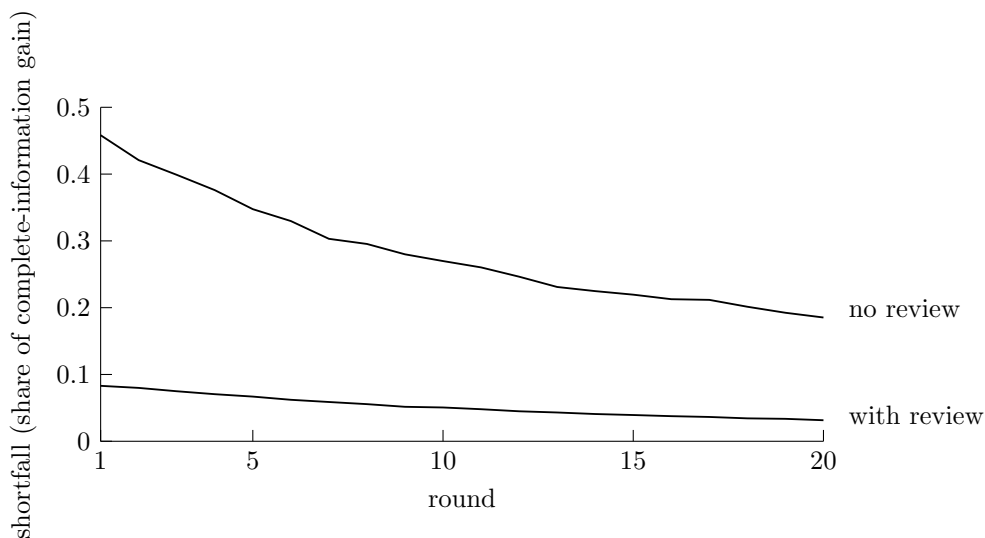


FIGURE 4. Records do not catch up. The shortfall of each funder’s expected research output below the complete-information optimum, as a share of what complete-information funding adds, round by round in the intermediate field. The no-review funder’s shortfall falls from 46 percent in round one to 19 percent by round twenty; the funder with a single review score observed at the start is at 8 percent in round one and 3 percent by round twenty. Settings: Appendix F.

whom to fund has little left to add: review is worth most where the budget is small relative to researchers’ resources. This ordering of review’s value by field does not depend on our choice of production function (Appendix C), and it survives the checks a skeptic would ask for: it holds when review is compared across fields at equal rank correlation between score and capability rather than at equal noise, for applicant pools of 25 to 200, for heavier and lighter tails of the resource distribution, when the review score partly reflects resources rather than capability alone, when review noise is multiplicative rather than additive, when review is repeated afresh every round rather than observed once, and across resource signals from nearly exact to nearly uninformative (Appendix D). Of these variants, multiplicative review noise, a constant relative error, is the one that narrows the heavy-tailed field’s advantage, because a constant relative error is a large absolute error for the researchers who matter most there: at equal informativeness review recovers 0.66 of the shortfall in the heavy-tailed field against 0.49 in the evenly spread field, where the additive signal gives 0.87 against 0.55. What review supplies, in every one of these variants, is information about capability that research output cannot supply on its own; peer review is one source of such information, and the results apply to any other, such as a pilot grant large enough to make

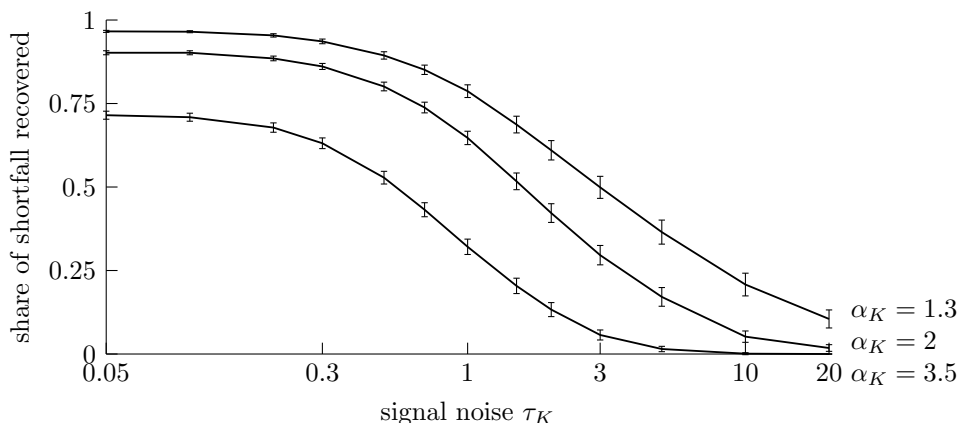


FIGURE 5. Review is worth most where capability is heavy-tailed. The share of the no-review shortfall that the review signal recovers, as review noise τ_K increases, for three capability distributions with the same mean capability (tail parameters $\alpha_K = 1.3, 2$, and 3.5 ; mean 2 in each). Bars: one standard error over 100 simulated populations. Settings: Appendix F.

research output informative (§5). One feature changes how much any of this is worth: the association between capability and resources. Where the capable already hold the resources, the gaps are small and even, and both targeting and review are worth less; where resources sit with the less capable, both are worth more.

A consequence concerns where review effort should go. In our model, what adds research output is separating the few researchers of unusually high capability from the rest, and that separation requires little information; fine distinctions among the many applications of comparable merit add little research output. A review system that spends most of its effort on those fine distinctions may be misallocating that effort.

These results bear on the dispute over peer review. Among funded NIH grants, percentile scores predict subsequent productivity barely better than chance; in our model, a signal of that accuracy is very noisy review.¹⁰ Whether such review is worth having depends on the field: at the noisiest signal in our sweep, review still recovers about a tenth of the no-review shortfall under heavy-tailed capability,

¹⁰ An AUC of 0.54 (Fang et al. 2016) corresponds to $\tau_K \gtrsim 20$ at the default parameters, the noisiest signal in our sweep. The mapping depends on the field: at the same noise, measured accuracy is higher under heavy-tailed capability, because the most capable researchers stand further apart.

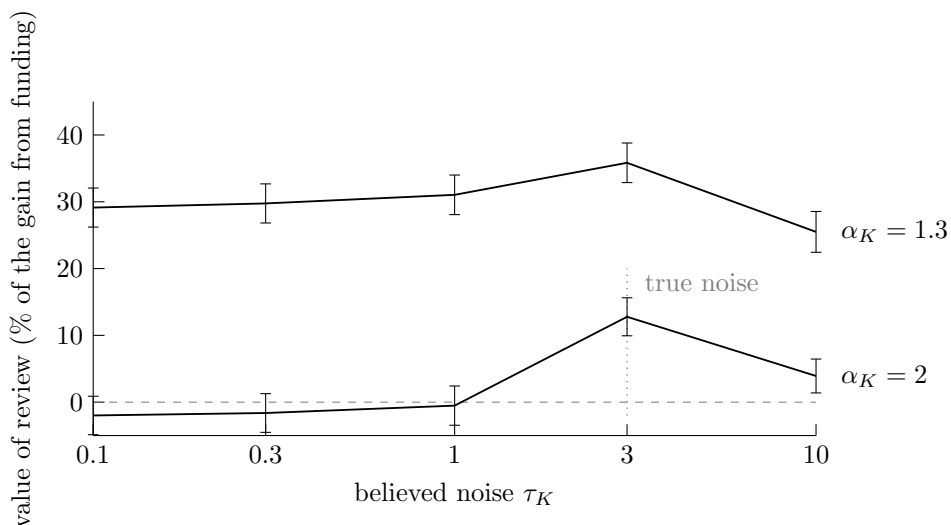


FIGURE 6. Overtrusting review loses more research output than undertrusting it. The value of review as a percent of the research output that a correctly calibrated review-informed funder adds over no funding, as the funder’s believed signal noise varies while the true noise is $\tau_K = 3$ (dotted rule); bars are one standard error. At the intermediate capability distribution, overtrust takes the value below zero. Settings: Appendix F.

about a fiftieth in the intermediate field, and none where capability is spread more evenly.¹¹ No single number settles the dispute; §7 returns to it.

One condition qualifies these results: the funder must weight review at its true informativeness. A funder that overtrusts review, treating a noisy signal as reliable, can do worse than one that ignores review altogether: with moderately spread capability, overtrust turns review’s value negative; with heavy-tailed capability it reduces the value without reversing it (Figure 6).¹² Overtrusting review loses more research output than undertrusting it.

Review can inform whom to fund. Next we turn to two alternatives to targeting that recur in funding policy, uniform seed grants and lotteries, and to how widely funds should be spread.

¹¹ 10.5, 1.8, and 0.0 percent of the no-review shortfall at $\tau_K = 20$, for $\alpha_K = 1.3$, 2, and 3.5, with the same mean capability.

¹² A funder that overtrusts tenfold (true $\tau_K = 3$, belief 0.3) loses more than review’s full value at the intermediate capability distribution; a funder that undertrusts tenfold (true $\tau_K = 0.3$, belief 3) retains over half of it.

6. SPREADING FUNDS: SEED GRANTS, LOTTERIES, AND CONCENTRATION

Targeting sends each dollar to the researcher expected to produce the most with it. Two alternatives to targeting recur in funding policy, and each has a rationale. Uniform seed grants give every researcher the same amount, so that a researcher whom review misjudges still receives something; in practice a funder gives out part of its budget as seed grants and targets the rest. Lotteries give grants to applicants chosen at random, usually from among those who pass a review threshold; their advocates argue that review cannot reliably rank the applicants it deems fundable, that review is costly, and that review is biased (§2). Both alternatives give up part of targeting on purpose. Our model can say how much expected research output is lost by giving up targeting, field by field, and where the loss is small. What the alternatives save, in the cost of review and in the effort researchers spend on proposals, lies outside our model and has been studied by others (§2).

Both alternatives lose the same thing: the value of targeting, the additional expected research output that the optimal allocation produces over an equal division of the same budget. How much research output is lost by giving up targeting depends on the field. For a funder allocating with records and review: in a heavy-tailed field with reliable review and a tight budget, targeting adds more research output than uniform funding’s entire gain over no funding; in the intermediate field at the largest budget in our parameter sweep, targeting adds about a quarter of that gain.¹³ Giving up targeting loses little research output under the same conditions that make review worth little: where capability is evenly spread, the budget is large, or review is uninformative.

Consider uniform seed grants first. The research output lost increases faster than the fraction of the budget given out as seed grants: giving out half of the budget as seed grants loses more than twice the research output that giving out a quarter loses (Figure 7). The size of the loss is set by what targeting is worth in the field. When a review-informed funder gives out three quarters of its budget as seed grants, it forfeits about a tenth of the research output its funding adds in a heavy-tailed field with reliable review, more at the smallest budget in our parameter sweep, and a few percent in the intermediate field.¹⁴ When the whole budget is given out as seed grants, the scheme is uniform funding, and the full

¹³ The review-informed funder’s gain over uniform funding, as a fraction of uniform funding’s gain over no funding, decreases from 1.17 to 0.72 across budget scales $b = 0.2$ to 2 with heavy-tailed capability and reliable review ($\tau_K = 0.3$), and from 0.74 to 0.25 in the intermediate field with the default signal.

¹⁴ About 11 percent, 17 percent at the smallest budget in our parameter sweep, and 3.5 percent in the intermediate field, as shares of the research output that the review-informed funder’s grants add over no funding.

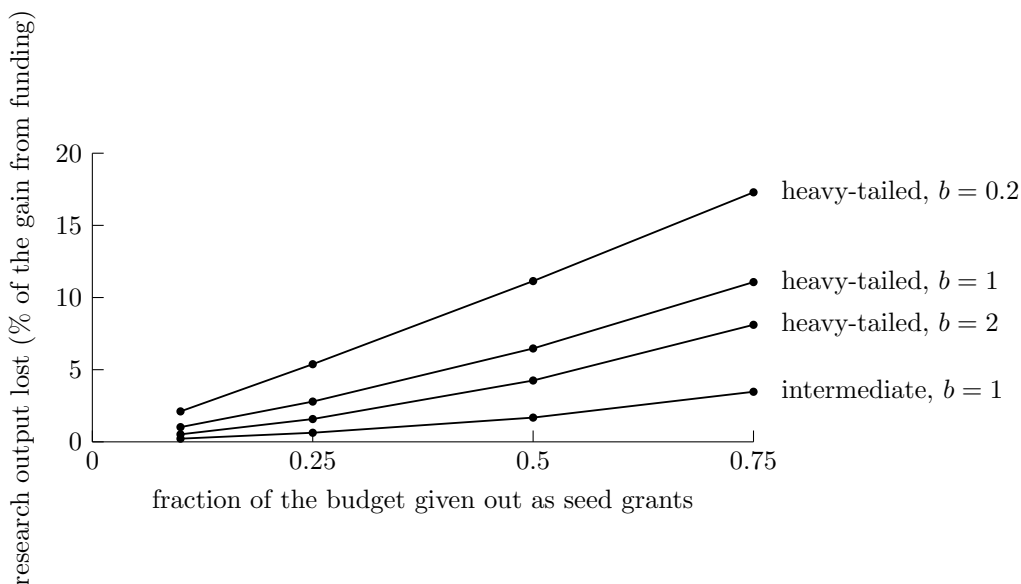


FIGURE 7. The research output lost to uniform seed grants. Expected research output lost, as a percent of the research output that the funder’s grants add over no funding, when a review-informed funder gives out a fraction of its budget as equal grants to all researchers and targets the rest. The loss grows faster than the fraction, and its size is set by what targeting is worth in the field. Settings: Appendix F.

value of targeting is lost. Whether the reasons for seed grants justify the loss is a judgment our model does not make; the model supplies the size of the loss.

Now consider lotteries. The lotteries funders run are partial: review sets a fundable pool, and chance picks the grantees within it. Such a lottery gives up two things, the review ranking within the pool and the certainty of who is funded, and the question is what each costs. We evaluate five schemes in a single round (Figure 8): equal division among the fifth of researchers with the highest review scores (selection by review); equal division among the top two fifths; the partial lottery that some funders run, which funds half of the top two fifths at random with full-size grants; uniform funding; and a lottery over all researchers with full-size grants. In a heavy-tailed field with a tight budget, selection by review gains about three quarters of what the optimal allocation gains when review is reliable and about half when review is very noisy. The partial lottery gains less, and most of the difference comes from ignoring the review ranking within the top two fifths rather than from the randomness itself.¹⁵ The lottery over all researchers gains

¹⁵ At review noise $\tau = 1$: selection by review gains 0.75 of what the optimal allocation gains, the equal division among the top two fifths 0.61, and the partial lottery 0.56; of the partial lottery’s 0.18 shortfall, ignoring the ranking accounts for 0.14 and the randomness for 0.04.

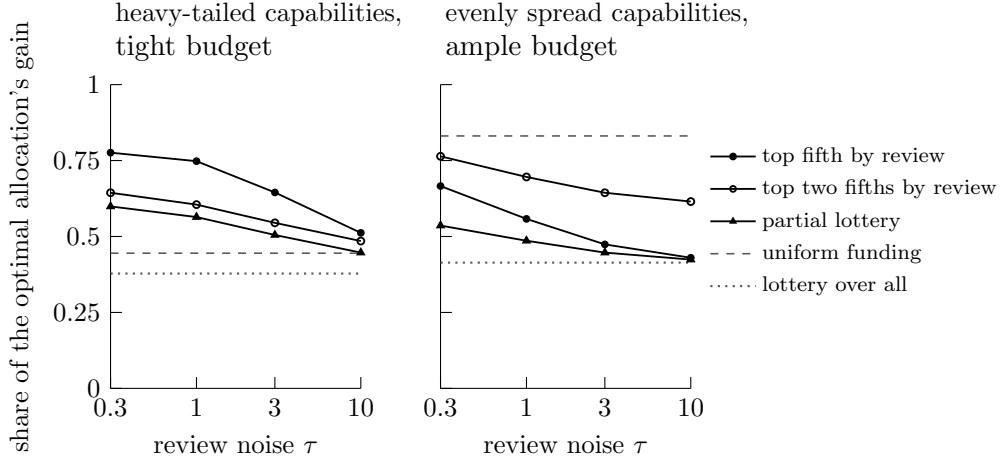


FIGURE 8. What five funding schemes gain, by field. Each scheme's gain over no funding as a share of the optimal allocation's gain, as review noise increases, in a heavy-tailed field with a tight budget (left) and an evenly spread field with a large budget (right). The top fifth and top two fifths by review score receive equal division; the partial lottery funds half of the top two fifths at random with full-size grants. Settings: Appendix F.

least, less than uniform funding. In an evenly spread field with a large budget, the ordering reverses: uniform funding gains more than four fifths of what the optimal allocation gains, and every scheme that concentrates grants, whether by review score or by chance, gains less.

A lottery among applicants above a review threshold therefore loses little research output only where the review ranking it ignores is worth little: replacing that ranking by chance loses a few hundredths of the optimal allocation's gain in an evenly spread field or when review is very noisy, and about a sixth of that gain in a heavy-tailed field with reliable review.¹⁶ The lottery debate rarely asks whether to concentrate grants at all. In an evenly spread field, our model's answer is to spread the budget in small grants across many researchers; a lottery among a few applicants above a review threshold produces less. Whether a lottery's savings in review costs and proposal effort outweigh the research output it loses depends on both quantities; the contest models cited in §2 measure the savings, and our model measures the loss.

The randomness itself costs research output for a general reason. A lottery gives each applicant in its pool a chance at a full grant instead of a certain smaller one,

¹⁶ The partial lottery's shortfall relative to selection by review: 0.01 to 0.07 of the optimal allocation's gain in the evenly spread field and at the noisiest review in the heavy-tailed field; 0.14 to 0.18 in the heavy-tailed field with reliable to moderately noisy review.

and on average the applicant receives the same amount either way; because expected research output increases with resources at a diminishing rate, the certain grant produces more. Dividing a pool’s money equally among the pool therefore produces more expected research output than any lottery over the same pool, provided that grants are divisible and that research output rises with resources at a diminishing rate at every grant size (Proposition 2, Appendix B). Where a project needs a minimum viable grant, the first condition fails and a lottery over full awards can beat dividing the same money into grants too small to use (Appendix B). The cost of the randomness is small in our simulations; the cost of the discarded ranking is the one that depends on the field.

Seed grants and lotteries both spread funds more widely than targeting does, and both raise the older question of whether funding should be concentrated among a few researchers or spread across many. One might hope that the production function settles this question: the harder it is to substitute resources for capability, the more the money should go to the few researchers who can use it. To test this, we embed our production function in a family indexed by an exponent γ , from Cobb-Douglas ($\gamma = 0$), under which a shortfall in either input can be offset by more of the other, through our form ($\gamma = -1$), to Leontief, under which research output depends only on the scarcer input; the harder it is to substitute one input for the other, the stronger the complementarity (Appendix C). The production function does not settle the question (Figure 9). With a tight budget in a heavy-tailed field, stronger complementarity does concentrate the review-informed funder’s grants. With an ample budget, stronger complementarity spreads them, a pattern consistent with a funder that, once the budget allows, brings every researcher’s resources up toward the level their capability can use. Where capability is evenly spread and the budget ample, the relation has no single direction: concentration peaks at an intermediate complementarity and declines toward both ends.¹⁷ How widely funds should be spread is set jointly by the production function, the budget, and the field’s capability distribution; no one of the three decides it alone.

Next we take stock: what the model implies for funders, and what its assumptions leave out.

¹⁷ Gini coefficients of the review-informed funder’s grants, from Cobb-Douglas to Leontief: 0.92 to 0.95 in a heavy-tailed field with a tight budget; 0.35 to 0.21 in the intermediate field with an ample budget; 0.63 to 0.45 in a heavy-tailed field with an ample budget. In the evenly spread field with an ample budget the Gini coefficient is 0.16 at Cobb-Douglas, peaks at 0.196 near $\gamma = -6$, and falls to 0.185 at $\gamma = -12$ (3.5 standard errors above the $\gamma = -12$ level).

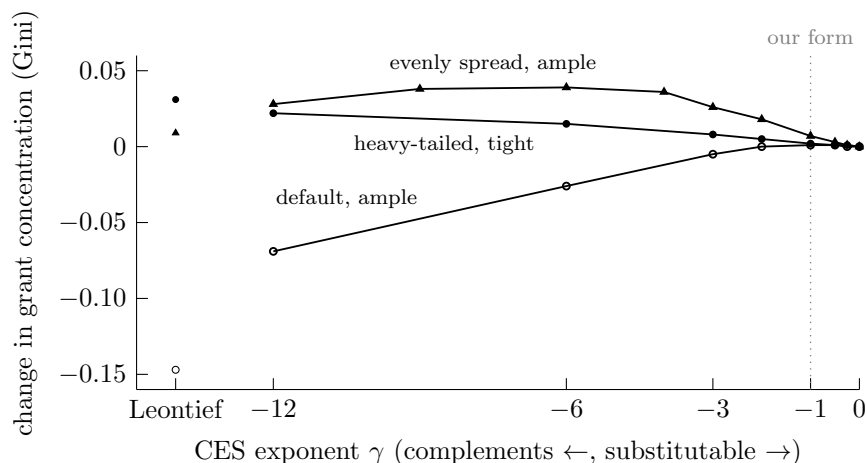


FIGURE 9. Complementarity concentrates funding only when the budget is small. The change in the Gini coefficient of the review-informed funder’s grants, relative to its Cobb-Douglas level, as the production function moves from Cobb-Douglas ($\gamma = 0$) toward Leontief (detached marks at the left edge), for three field-and-budget combinations. Settings: Appendix F.

7. DISCUSSION

We have compared the expected impact on research output of targeted funding to what it would be under equal division of the same budget, and to what it would be in the absence of any contribution by the funder. Our results isolate one quantity as key to strategies for optimal grant funding: the value of targeting researchers with the greatest gap between their capabilities and their resources. Two features of an area of research determine this value. The first is the funder’s budget relative to researchers’ own resources. Where the budget is small, dollars are scarce and where they go matters; where the budget can relieve most researchers’ bottlenecks, the optimal allocation and equal division produce nearly the same research output (§4, Figure 2). This saturation holds when capability bounds research output, as under the harmonic mean; under a production function with no such bound, such as Cobb-Douglas, the gap rule still holds but the value of targeting need not vanish as the budget grows (Appendix C). The second is how unequally the capability-resource gaps are distributed. Across every setting we considered, the value of targeting is close to a function of the inequality of the optimal grants alone (Appendix D), and what makes the gaps unequal is mainly how unequal capability is: where a few researchers can produce far more than the rest, getting resources to those few is most of what a funder can add (§5). The association between capability and resources works through the same channel: where the capable already hold the resources, the gaps are small and even, and targeting is

worth less. Two results are not set by this quantity alone. How widely a review-informed funder should spread its grants depends, in addition, on how strongly capability and resources complement each other (§6, Figure 9); and the timing of spending over the horizon, which Appendix E analyzes, adds little research output beside the choice of recipients in every setting we test, subject to the limits stated there.

We showed how much various sources of information contribute to the effectiveness of targeting this gap. Track records are modestly predictive of future research output, but they do not pin down whether a researcher’s productivity is bottlenecked by resources or by capability, and further rounds of records improve the funder’s allocation only slowly (§5). Peer review of a proposal can provide a signal of capability less confounded by resources, and the value of that signal increases with the two features that set the value of targeting and with the signal’s informativeness; where capability is heavy-tailed, even moderately noisy review recovers most of that value (§5). Large grants make records informative, and spending later lets the funder allocate with more information, but the gain from planning ahead is smaller than the gain from the review signal at every horizon and compounding rate in our parameter sweep (Appendix E). We also measured what a funder loses by forgoing targeting: seed grants and lotteries lose most research output where the value of targeting is largest, most of what a partial lottery loses comes from discarding the review ranking within its pool rather than from chance, and whether funding should be concentrated or spread depends on the budget and the field, not on the production function alone (§6).

These results have a practical consequence for funders. Within the model, no funding mechanism is best in every field. Peer review, seed grants, and lotteries each perform better in some fields than in others, and the two features of the field decide which. A funder choosing among mechanisms should therefore begin by locating its field on the two (Figure 10). The first feature is observable. A funder’s budget relative to the field’s own resources can be read from administrative data: its annual awards to the field against the field’s research spending from every other source. The second feature is not. The inequality of the gaps depends on capabilities the funder cannot see and on the budget itself, and the inequality of the optimal grants, the quantity that organizes our results (Appendix D), is not one a funder can compute. Three observable quantities stand in for it. The inequality of research output in the field is the natural stand-in for the inequality of capability, though where grants are small research output tracks resources rather than capability (§5), so this stand-in leans toward the resource distribution. The dispersion of review scores over the whole applicant pool is a second,

with the same caveat about noise. The most direct is the response of research output to grants large enough to relieve the bottleneck: on a grant comparable to the recipient’s capability, research output becomes informative about capability (Figure 3), so a program of substantial pilot grants measures the gaps, whereas small seed grants do not, because on small grants research output tracks resources for every recipient. How existing funding is distributed completes the estimate: where resources already sit with the capable, the gaps are smaller and more even than capability inequality alone would imply.

Figure 10 places four kinds of funding program on the budget axis by order of magnitude, from 2023 and 2024 budget figures; a round corresponds to a year of awards and the map’s budget covers two rounds, so a program whose annual awards are a fraction r of its field’s other annual resources sits at $b = 2r$ (Appendix F). A national biomedical agency that supplies a large share of its field’s research support, with annual awards comparable to the field’s other resources, sits at the top of the range; a national science agency that funds its fields more thinly sits near a tenth of that; a continental excellence council near a few hundredths; and a private foundation funding one field among many at a hundredth or below. Where a field lies on the inequality axis is the estimate each funder must make, and the range of tail parameters estimated for research productivity spans most of the axis, so the placements say which regime a funder’s budget puts it in, not where it sits within the regime. The regimes differ. A foundation or council with a small budget in a heavy-tailed field faces the largest value of targeting per dollar: review directed at identifying the few researchers of unusually high capability is worth most, even moderately noisy review recovers most of that value, seed grants give up the most research output, and a lottery among applicants above a review threshold gives up a substantial share of what selection by review gains. An agency whose budget approaches the field’s other resources, in a field where capability is evenly spread, faces a small value of targeting: review of any informativeness adds little, seed grants and lotteries give up little, and the model’s best use of the budget is small grants to many researchers.

Three further implications concern peer review. The first concerns where review effort should go. Most of review’s value comes from separating the few researchers of unusually high capability from the rest, and fine distinctions among the many applications of comparable merit add little research output. This suggests that a review system that spends most of its effort ranking the middle of the distribution spends that effort where the model locates the least research output (§5). The second concerns how to read the evidence on review’s efficacy. The finding that percentile scores barely predict productivity comes from funded NIH grants

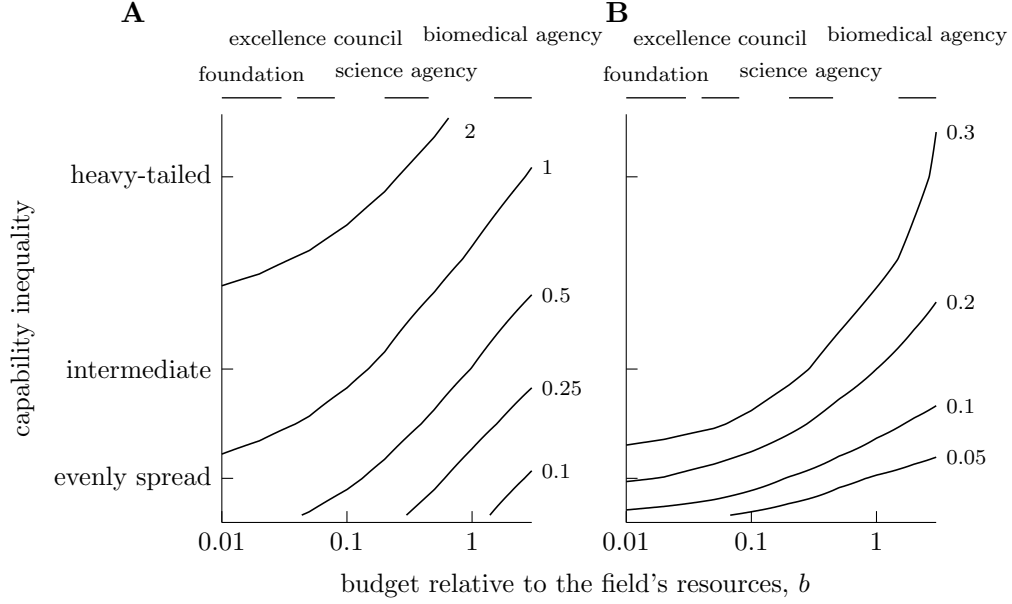


FIGURE 10. Where a funder stands. (A) The value of targeting: the complete-information optimum's gain over uniform funding, as a share of uniform funding's gain over no funding. (B) The value of review at the default noise, as a share of the research output that review-informed funding adds. Both by the funder's budget relative to the field's resources and the inequality of capability, at fixed mean capability; the vertical axis is the Gini coefficient of capability, and its ticks mark the three fields of the paper ($\alpha_K = 1.3, 2$, and 3.5); each contour is labeled by its value. The bars above each panel place four kinds of program on the budget axis by order of magnitude, from 2023 and 2024 annual budget figures with b equal to twice the annual ratio, since the map's budget covers two rounds (Appendix F); where a field lies on the vertical axis is the estimate each funder must make, and the tail parameters estimated for research productivity lie in its upper part. Two rounds, $\tau_K = 1$, 50 populations per cell; contours interpolated between the 63 cells. Settings: Appendix F.

with scores in the top fifth (Fang et al. 2016), and the studies that find modest predictive power also use funded grants (Li and Agha 2015). Both kinds of study measure prediction among applicants that review has already selected, that is, after review has made the separation the model values most, and within the range of applicants where the model expects review to add least. A weak association within that band is consistent with review of considerable value over the whole applicant pool, as the statistical analysis of these studies by Lindner and Nakamura (2015) also concludes. For the same reason, the mapping in §5 from measured accuracy to signal noise, which treated the measured accuracy as review's accuracy over the

whole pool, understates review’s informativeness. The third concerns how much a funder should trust review. A funder that treats review as more reliable than it is loses more research output than a funder that treats review as noisier than it is (§5, Figure 6). A funder uncertain about review’s accuracy should therefore err toward undertrust.

The model holds several things fixed, and its results hold within those limits. There is one funder. Competition among funders, and the process by which past funding attracts future funding across funders, lie outside the model, though the gap rule itself directs grants away from researchers whose resources are already large. Researchers do not respond strategically. Proposal effort, risk choice, and the cost of review lie outside the model; the contest models cited in §2 cover that ground, and the mechanism-design treatment of grants by Carnehl et al. (2025), in which the funder must also elicit information from applicants who bear the cost of applying, is the natural complement to a model in which the funder’s information arrives for free. Research output is a single number per round. Projects do not differ in kind, and exploring new directions has no value in the model, so results about funding unexplored regions of a field have no counterpart here; nor do a researcher’s proposals differ in quality, so this is a model of investigator-level funding, and project-specific quality that review might observe is the natural extension. Grants are divisible, so a minimum viable grant size, which can favor a lottery over full awards, is outside the model (Appendix B). The review signal is noisy but unbiased. Bias and conservatism in review, which are among the strongest arguments for lotteries, are not modeled (Ginther et al. 2011; Wittteman et al. 2019; Boudreau et al. 2016). Researchers do not enter or leave the field, and have no careers. Capability is a state rather than a trait, so it holds whatever a researcher has accumulated that raises research output, and a funder that reads it as innate ability misreads the model. The asymmetry between the two inputs is an assumption with consequences: capability persists and grows, resources do not persist, so a grant’s lasting effect enters only through capability. Durable equipment, data, and infrastructure violate this asymmetry. The one-round results do not depend on it; the results about compounding and about the timing of spending (Appendix E) do. The compounding here is of capability through doing research, and differs from the cumulative advantage in which winning a grant raises the chance of winning the next (Merton 1968; Bol et al. 2018); the two would need to be separated in any empirical test. Inputs are complements, and the timing results depend on complementarity while the information results do not (Appendix C). The funder is an idealization that knows the model and updates correctly, so the values of information reported here are what such a funder extracts, not what a real funder would extract. Finally, the numbers we report are comparisons across

the parameter values in our sweep, and none is calibrated to a real funder. What applies to a real field is the qualitative features of the dynamics that organize the results.

The model makes four predictions that data on funded research can test. The effect of winning a grant on research output should be larger for recipients whose other resources are small relative to their capability, so a randomized trial of funding, such as the one that found no clear average effect (Barnett et al. 2024), should find its effect concentrated among under-resourced recipients, and a null average is what the model predicts where most recipients are already well resourced. The remaining three use data that funders already hold. First, measures that stand in for capability, such as prior record or review score, should predict research output better the larger a grant is relative to the recipient’s other resources, because on small grants research output is approximately proportional to resources and nearly independent of capability (§5, Figure 3). A test compares the predictive power of review scores across grant sizes within one applicant pool. Second, how well review scores predict research output, measured over the whole applicant pool rather than among funded grants, should be higher in fields whose research output distributions have heavier tails (§5). A test compares fields, or one funder’s programs, using the full range of scores where the outcomes of funded and unfunded applicants can both be observed. Third, where funders run lotteries beside review, the research output lost to the lottery should be larger in heavier-tailed fields (§6, Figure 8). The lottery programs now running at several funders supply the comparison. The second and third tests compare fields at similar budget scales and review informativeness, and read the tail of research output as a stand-in for the tail of capability. The three tests need grant sizes and recipients’ other funding, review scores and outcomes over whole applicant pools, and the research output of lottery-funded and review-funded grants where a funder runs both.

8. CONCLUSION

Our model provides a unifying insight into the grant funder’s challenge. The output-maximizing allocation funds the gap between each researcher’s capability and their resources, and neither of the intuitive rules, funding the track record or funding the under-resourced, does so, because each tracks only one of the two. What the funder can learn about the gap sets everything else. Records alone do not pin it down, because research output does not show which of the two is scarce; a signal of capability less confounded by resources does, and peer review can provide one. What that signal is worth and what seed grants and lotteries lose turn on the same two features of a field: how much of researchers’ resources the funder’s budget can supply, and how unequal the capability-resource gaps are,

which is mostly a matter of how unequal capability is. Where the budget is tight and capability heavy-tailed, targeting is worth a great deal, even moderately noisy review captures most of that value, and giving up targeting loses much of it. Where the budget is ample and capability evenly spread, little research output is at stake and spreading funds widely loses almost nothing. The map of the two features has regimes in which the answers differ, and real funders occupy different regimes. A funder deciding among review, lotteries, and seed grants should therefore begin by asking which kind of field it is funding.

Appendices

A. DERIVATION OF THE GAP RULE

The within-round allocation problem. A population of n researchers. Researcher i has capability $K_i > 0$ and baseline resources $R_{i0} \geq 0$, both fixed within the round. A grant $g_i \geq 0$ adds to researcher i 's resources for the round, so that their expected research output is $F(K_i, R_{i0} + g_i)$. We prove the gap rule for every production function of the form

$$F(K, R) = A K h(R/K),$$

where $A > 0$ is the productivity constant and $h: [0, \infty) \rightarrow [0, \infty)$ is continuous, continuously differentiable on $(0, \infty)$, strictly increasing, and strictly concave, so that h' is positive and strictly decreasing. This is the class of production functions with constant returns to scale in (K, R) and diminishing returns to resources. The harmonic mean of §3 is the instance $h(x) = x/(1+x)$, for which $F(K, R) = AKR/(K+R)$; every member of the power-mean family of Appendix C with exponent $\gamma < 1$ is an instance, with $h(x) = ((1+x^\gamma)/2)^{1/\gamma}$. Write

$$\varphi_i(g) = A K_i h\left(\frac{R_{i0} + g}{K_i}\right)$$

for researcher i 's expected research output as a function of their grant alone; for the harmonic mean, $\varphi_i(g) = AK_i(R_{i0} + g)/(K_i + R_{i0} + g)$.

Throughout, the round's budget is $B > 0$. Nothing else about the model is used, so the result holds for every realized population and every round's budget; across rounds, the constant c and the funded set are recomputed from the new population state.

The funder observes $(K_i, R_{i0})_{i=1}^n$ and divides the budget so as to maximize expected research output:

$$(P) \quad \max_{g_1, \dots, g_n \geq 0} \sum_{i=1}^n \varphi_i(g_i) \quad \text{subject to} \quad \sum_{i=1}^n g_i \leq B.$$

Three lemmas.

Lemma 1 (the return to a grant). *For each i , on $g \in [0, \infty)$: (i) φ_i is strictly increasing, with $\varphi'_i(g) = A h'((R_{i0} + g)/K_i) > 0$; (ii) φ_i is strictly concave, since φ'_i is strictly decreasing in g ; (iii) $\varphi_i(g) \uparrow AK_i \bar{h}$ as $g \rightarrow \infty$, where $\bar{h} = \lim_{x \rightarrow \infty} h(x) \leq \infty$. For the harmonic mean, $\varphi'_i(g) = AK_i^2/(K_i + R_{i0} + g)^2$, $\varphi''_i(g) = -2AK_i^2/(K_i + R_{i0} + g)^3$, $\varphi'_i(g) < A$ whenever $R_{i0} + g > 0$, and $\bar{h} = 1$, so expected research output never exceeds AK_i .*

Proof. Differentiating $\varphi_i(g) = AK_i h((R_{i0} + g)/K_i)$ in g gives (i), and (ii) follows because h' is strictly decreasing and its argument increases with g . For (iii), h is increasing, so $h(x) \uparrow \bar{h}$ as $x \rightarrow \infty$. For the harmonic mean, write $S = K_i + R_{i0} + g$; then $\varphi_i(g) = AK_i - AK_i^2/S$, and the stated derivatives follow (note $dS/dg = 1$); $\varphi'_i(g) = AK_i^2/S^2 < A$ exactly when $S > K_i$, that is, when $R_{i0} + g > 0$. $\square$

Lemma 1 says that the marginal value of a dollar to researcher i depends on capability and resources only through their ratio, decreases as that ratio rises, and so increases with capability and decreases with the resources they already hold; that each dollar granted lowers the value of the next; and that, when h is bounded, expected research output never exceeds a ceiling set by capability.

Lemma 2 (existence, uniqueness, exhaustion). *Problem (P) has exactly one solution $g^* = (g_1^*, \dots, g_n^*)$, and it exhausts the budget: $\sum_i g_i^* = B$.*

Proof. The feasible set is compact and convex and the objective $f(g) = \sum_i \varphi_i(g_i)$ is continuous and, by lemma 1(ii), strictly concave, so a maximizer exists and is unique. If $\sum_i g_i^* < B$, increasing any one g_i^* slightly stays feasible and raises f (lemma 1(i)), contradicting optimality. $\square$

Lemma 3 (equal marginal value). *A feasible allocation g with $\sum_i g_i = B$ solves (P) if and only if there is a $\nu > 0$ such that*

$$\varphi'_i(g_i) = \nu \quad \text{for every } i \text{ with } g_i > 0, \quad \varphi'_i(0) \leq \nu \quad \text{for every } i \text{ with } g_i = 0,$$

where $\varphi'_i(0)$ may be $+\infty$ (as it is when $R_{i0} = 0$ and $h'(0^+) = \infty$), in which case researcher i is funded.

Proof. Necessity. Let g^* solve (P). Since $B > 0$ is exhausted (lemma 2), some researcher is funded: $g_i^* > 0$ for some i . Fix any such i and any $j \neq i$, and for $0 < \delta \leq g_i^*$ consider the transfer that replaces g_i^* by $g_i^* - \delta$ and g_j^* by $g_j^* + \delta$, leaving the rest unchanged. The result is feasible, so optimality gives

$$\varphi_j(g_j^* + \delta) - \varphi_j(g_j^*) \leq \varphi_i(g_i^*) - \varphi_i(g_i^* - \delta).$$

Dividing by δ and letting $\delta \downarrow 0$ yields $\varphi_j'(g_j^*) \leq \varphi_i'(g_i^*)$. If j is also funded, the same argument with i and j exchanged gives the reverse inequality, so all funded researchers share a common marginal value; call it $\nu = \varphi_i'(g_i^*) > 0$ (positivity by lemma 1(i)). For unfunded j , the displayed inequality reads $\varphi_j'(0) \leq \nu$.

Sufficiency. Let g satisfy the condition with multiplier ν , and let $\tilde{g}$ be any feasible allocation. Concavity of φ_i gives, for every i , $\varphi_i(\tilde{g}_i) \leq \varphi_i(g_i) + \varphi_i'(g_i)(\tilde{g}_i - g_i)$. For funded i , $\varphi_i'(g_i) = \nu$, so $\varphi_i'(g_i)(\tilde{g}_i - g_i) = \nu(\tilde{g}_i - g_i)$. For unfunded i , $g_i = 0$ and $\tilde{g}_i - g_i = \tilde{g}_i \geq 0$, so $\varphi_i'(0)(\tilde{g}_i - g_i) \leq \nu(\tilde{g}_i - g_i)$. Summing over i :

$$f(\tilde{g}) \leq f(g) + \nu \sum_i (\tilde{g}_i - g_i) = f(g) + \nu \left(\sum_i \tilde{g}_i - B \right) \leq f(g),$$

since $\tilde{g}$ is feasible and $\nu > 0$. So g solves (P). $\square$

Lemma 3 is the formal content of the sentence in § 4: an allocation is optimal exactly when no dollar can be moved to a researcher who can produce more value with it.

The proposition.

Proposition 1 (the gap rule). *Let $F(K, R) = A K h(R/K)$ with h increasing and strictly concave. Then problem (P) has the unique solution*

$$g_i^* = \max(c K_i - R_{i0}, 0), \quad i = 1, \dots, n,$$

where $c > 0$ is the unique constant satisfying the budget identity $\sum_{i=1}^n \max(c K_i - R_{i0}, 0) = B$. Equivalently, $c = (h')^{-1}(\nu/A)$, where ν is the common marginal value of lemma 3; for the harmonic mean, $c = \sqrt{A/\nu} - 1$ with $\nu \in (0, A)$.

Proof. Let g^* be the unique solution (lemma 2) and ν the multiplier that lemma 3 associates with it. The proof proceeds in four steps.

Step 1: $c = (h')^{-1}(\nu/A)$ is well defined and positive. Some researcher i is funded, and for them lemma 3 gives $\nu = \varphi_i'(g_i^*) = A h'((R_{i0} + g_i^*)/K_i)$, so ν/A lies in the range of h' ; since h' is strictly decreasing, its inverse is defined there, and $c = (h')^{-1}(\nu/A) = (R_{i0} + g_i^*)/K_i > 0$ because $g_i^* > 0$. For the harmonic mean, $h'(x) = 1/(1+x)^2$, so $(h')^{-1}(y) = 1/\sqrt{y} - 1$ and $c = \sqrt{A/\nu} - 1$; positivity of c is then $\nu < A$, which lemma 1 gives directly.

Step 2: funded researchers receive exactly their gap. Let $g_i^* > 0$. By lemma 3, $A h'((R_{i0} + g_i^*)/K_i) = \nu$, so $(R_{i0} + g_i^*)/K_i = (h')^{-1}(\nu/A) = c$ and $g_i^* = c K_i - R_{i0}$. Since $g_i^* > 0$, this equals $\max(c K_i - R_{i0}, 0)$. Every funded researcher is thus brought to the same ratio of resources to capability, c .

Step 3: unfunded researchers have no gap. Let $g_i^* = 0$. By lemma 3, $A h'(R_{i0}/K_i) = \varphi'_i(0) \leq \nu = A h'(c)$, and since h' is strictly decreasing, $R_{i0}/K_i \geq c$, that is, $R_{i0} \geq c K_i$. Then $\max(c K_i - R_{i0}, 0) = 0 = g_i^*$.

Steps 2 and 3 establish $g_i^* = \max(c K_i - R_{i0}, 0)$ for every i , and the budget identity is exhaustion (lemma 2). It remains to show c is the only constant satisfying that identity.

Step 4: the budget identity pins c uniquely. Define $G(c) = \sum_{i=1}^n \max(c K_i - R_{i0}, 0)$ for $c \geq 0$. G is continuous and nondecreasing, with $G(0) = 0$ and $G(c) \rightarrow \infty$ as $c \rightarrow \infty$. Moreover G is strictly increasing wherever it is positive: if $G(c) > 0$, then $c K_j > R_{j0}$ for some j , and for any $c' > c$, $G(c') \geq G(c) + K_j (c' - c) > G(c)$. Now suppose $c_1 < c_2$ both satisfy $G(c) = B$. Since $B > 0$, $G(c_1) > 0$, so $G(c_2) > G(c_1) = B$, a contradiction. $\square$

The proof uses only that the marginal value of a dollar depends on K and R through the ratio R/K and falls as that ratio rises. Constant returns to scale give the first property and diminishing returns to resources the second; the harmonic form contributes nothing beyond the closed form of c . The constant c is the ratio of resources to capability at which the optimal allocation leaves every funded researcher, and so also the rate at which the rule converts a unit of capability into a target level of resources.

Consequences.

Corollary 1 (funding frontier and budget comparative statics). *(i) Researcher i is funded, $g_i^* > 0$, exactly when $K_i > R_{i0}/c$. (ii) As a function of the budget B , the constant $c(B)$ is continuous and strictly increasing; hence the funded set is nondecreasing in B , every funded researcher's grant $c(B) K_i - R_{i0}$ is strictly increasing in B , and the common marginal value $\nu(B) = A h'(c(B))$ is strictly decreasing in B . Both parts follow from the formula and Step 4.*

Corollary 2 (the value of targeting vanishes in the budget). *Let $f^*(B)$ denote the optimal value of (P) and $f^u(B) = \sum_i \varphi_i(B/n)$ the value of uniform funding. Then $0 \leq f^*(B) - f^u(B)$ for every B . If h is bounded, $\bar{h} < \infty$, then $f^*(B) - f^u(B) \rightarrow 0$ as $B \rightarrow \infty$.*

Proof. The first inequality holds because uniform funding is feasible. For the second: by lemma 1(iii), $\varphi_i(g) < AK_i\bar{h}$ for every g , so $f^*(B) \leq A\bar{h} \sum_i K_i$; and $f^u(B) = \sum_i \varphi_i(B/n) \rightarrow A\bar{h} \sum_i K_i$ as $B \rightarrow \infty$. Hence $0 \leq f^*(B) - f^u(B) \leq A\bar{h} \sum_i K_i - f^u(B) \rightarrow 0$. $\square$

Two remarks. First, the corollary says nothing about monotonicity: $f^*(B) - f^u(B)$ is zero at $B = 0$ and positive for small $B > 0$ whenever researchers differ, so it cannot decrease throughout. The decrease reported in the main text is a numerical finding about the ratio $(f^* - f^u)/(f^u - f^0)$, with f^0 the research output under no funding: it fell monotonically in the budget, on budget scales from 0.02 to 3, in every one of 300 populations of 50 researchers drawn from the default distributions, and in the population of Figure 2. Since uniform funding's own gain over no funding does not vanish (it approaches $A\bar{h} \sum_i K_i - \sum_i \varphi_i(0) > 0$), the ratio vanishes with the difference. Second, boundedness of h is what makes targeting's value vanish: in the power-mean family, h is bounded exactly when $\gamma < 0$, with $\bar{h} = 2^{-1/\gamma}$ (so $\bar{h} = 1$ for the harmonic mean), whereas under Cobb-Douglas production ($\gamma \rightarrow 0$, $h(x) = \sqrt{x}$) research output has no ceiling set by capability and $f^*(B) - f^u(B)$ grows without bound in B for any population with unequal capabilities. The Leontief form $h(x) = \min(1, x)$ is the limit $\gamma \rightarrow -\infty$ of the family; it is not strictly concave, so the proposition does not apply to it directly, and it is treated numerically in Appendix C.

Example 1 (funding the track record can produce less than uniform funding). *Two researchers, harmonic production, $A = 1$, budget $B = 2$. Researcher 1: $K_1 = 8$, $R_{10} = 8$ (productive and well resourced; expected research output $\varphi_1(0) = 4$). Researcher 2: $K_2 = 4$, $R_{20} = 1$ (capable but with few resources; $\varphi_2(0) = 4/5$). Allocating in proportion to expected research output gives $g = (5/3, 1/3)$ and total research output $285/53 \approx 5.38$. Uniform funding, $g = (1, 1)$, gives $284/51 \approx 5.57$. The gap rule ($c = 3/4$) gives $g^* = (0, 2)$ and $40/7 \approx 5.71$.*

Example 2 (funding the under-resourced can produce less than uniform funding). *Two researchers, harmonic production, $A = 1$, budget $B = 2$. Researcher 1: $K_1 = 10$, $R_{10} = 5$. Researcher 2: $K_2 = 1/2$, $R_{20} = 0$ (the scarcest resources in the population). Directing the budget to the researcher with the fewest resources gives total research output $56/15 \approx 3.73$. Uniform funding gives $49/12 \approx 4.08$. The gap rule ($c = 2/3$) gives $g^* = (5/3, 1/3)$ and $21/5 = 4.2$.*

The one-round rule and the sequential optimum. Proposition 1 maximizes one round's expected research output. Over T rounds a grant also raises the recipient's later capability, by $\epsilon \Lambda(K_i, R_{i0} + g_i)$ per round, so the allocation that maximizes expected research output over the horizon can differ from the gap rule in two

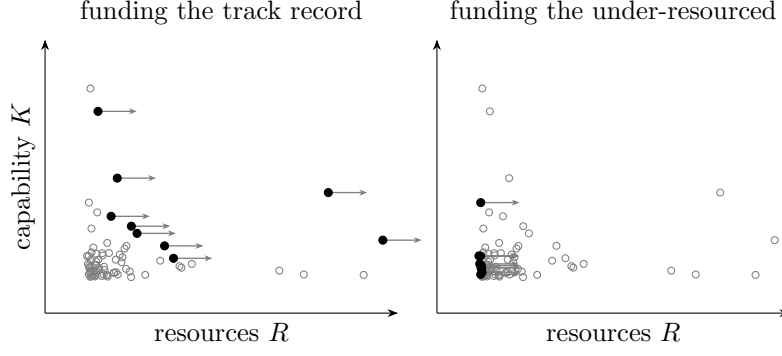


FIGURE 11. The two intuitive schemes on the population of Figure 1, with the budget of that figure’s smaller frontier divided equally among nine researchers. Left: funding the track record gives the grants (arrows) to the nine researchers with the highest expected research output; most of them are well resourced and of middling capability, so an added dollar adds little. Right: funding the under-resourced gives the grants to the nine researchers with the fewest resources, chosen without regard to capability, so most of the money goes to researchers of low capability. On this population the gap rule adds 4.0 units of expected research output over no funding; funding the track record adds 2.0, funding the under-resourced 2.1, and dividing the budget uniformly across all 75 researchers 2.0.

ways: in whom it funds within a round, since a dollar’s future value depends on the recipient’s growth, and in how it divides the budget across rounds. Because capability growth follows expected research output, the complete-information problem over T rounds is a deterministic program. For $T = 2$ we solve it to global optimality: the round-1 grants and the round-1 share of the budget are chosen freely (a softmax parameterization over the n recipients and a logistic share, optimized by quasi-Newton iteration from the gap rule as the starting point), and the remaining budget is allocated in round 2 by the gap rule on the grown capabilities, which is optimal for the last round. The two-round objective is concave in the grants, so the stationary point the iteration reaches is the global optimum up to numerical tolerance: the harmonic mean $\Lambda(K, R)$ is jointly concave and increasing in (K, R) ; the round-2 value of a capability vector, the maximum of $\sum_i \Lambda(K'_i, R_{i0} + g_{i2})$ over the round-2 grants, is therefore concave and increasing in K' , because partial maximization of a jointly concave function preserves concavity; and each grown capability $K'_i = K_i + \epsilon \Lambda(K_i, R_{i0} + g_{i1})/A$ is concave in g_{i1} , so the composition is concave in the round-1 grants. Table 2 compares this optimum with the benchmark used throughout the paper, the gap rule applied to half the budget in each round, on 20 populations of 50 at each setting, with mean capability 2. For $T = 5$

TABLE 2. The round-by-round gap rule against the sequential optimum under complete information. Two rounds: the research output lost by applying the gap rule to half the budget in each round, as a percent of the research output that the exact two-round optimum adds over no funding (mean and maximum over 20 populations); the correlation between the two allocations' first-round grants (mean and minimum); and the share of the budget the optimum spends in round one. Five rounds: the research output lost by an even division of the budget across rounds, as a percent of what the best division adds, with the gap rule choosing recipients in every round, and the center of mass of the best division (0.5 is even; higher is later). Mean capability 2; $n = 50$.

α_K	ϵ	b	two rounds, optimum			five rounds, division only	
			loss, % (max)	correlation (min)	round-1 share	even division loses, %	center of mass
1.3	0.1	0.2	0.0 (0.0)	1.000 (1.000)	0.49	0.0	0.51
1.3	0.1	1	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.51
1.3	0.1	2	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.51
1.3	0.5	0.2	0.1 (0.4)	0.999 (0.997)	0.46	0.4	0.54
1.3	0.5	1	0.2 (0.6)	1.000 (0.999)	0.45	1.0	0.55
1.3	0.5	2	0.3 (0.5)	1.000 (0.999)	0.45	1.4	0.55
1.3	0.85	0.2	0.3 (1.1)	0.999 (0.994)	0.44	0.9	0.56
1.3	0.85	1	0.6 (1.4)	0.999 (0.998)	0.42	2.1	0.57
1.3	0.85	2	0.7 (1.2)	1.000 (0.997)	0.42	2.9	0.58
2	0.1	0.2	0.0 (0.0)	1.000 (1.000)	0.49	0.0	0.51
2	0.1	1	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52
2	0.1	2	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52
2	0.5	0.2	0.2 (0.4)	0.999 (0.997)	0.45	0.6	0.55
2	0.5	1	0.3 (0.5)	1.000 (0.999)	0.44	1.2	0.57
2	0.5	2	0.3 (0.4)	1.000 (0.999)	0.44	1.6	0.57
2	0.85	0.2	0.4 (1.1)	0.997 (0.989)	0.41	1.2	0.58
2	0.85	1	0.8 (1.2)	0.999 (0.996)	0.41	2.6	0.60
2	0.85	2	0.8 (1.1)	0.999 (0.997)	0.41	3.3	0.60
3.5	0.1	0.2	0.0 (0.0)	1.000 (0.999)	0.48	0.1	0.52
3.5	0.1	1	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52
3.5	0.1	2	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52
3.5	0.5	0.2	0.3 (0.5)	0.997 (0.994)	0.41	1.3	0.60
3.5	0.5	1	0.4 (0.4)	0.999 (0.998)	0.43	1.8	0.59
3.5	0.5	2	0.4 (0.4)	0.999 (0.999)	0.44	1.9	0.58
3.5	0.85	0.2	0.8 (1.2)	0.993 (0.985)	0.36	2.8	0.65
3.5	0.85	1	0.9 (1.1)	0.997 (0.995)	0.39	3.7	0.63
3.5	0.85	2	0.9 (1.0)	0.998 (0.997)	0.41	4.0	0.62

we restrict the comparison to the division of the budget across rounds, with the gap rule choosing recipients in each round, and compare the best division with the even one.

The benchmark loses at most 1.4 percent of the research output that funding adds in any population, and under 1 percent on average in every setting; its grants correlate with the optimum's at 0.98 or above. The loss grows with the

compounding rate and with the budget, is larger where capability is evenly spread, and is indistinguishable from zero at the default compounding rate. The two-round optimum spends 36 to 49 percent of the budget in the first round, because the capability that round-1 research output builds makes round-2 dollars more productive; the same shift appears in the forward-looking funder of Appendix E. Over five rounds, with the gap rule choosing recipients, the even division of the budget across rounds loses at most 4.0 percent of what the best division adds at the highest compounding rate, and at most 0.1 percent at the default rate, and the best division shifts spending later (center of mass 0.51 to 0.65). The one-round rule is therefore the right benchmark for the sequential problem at the compounding rates we study, and the value of information reported in the main text is measured against a benchmark within about one percent of the sequential optimum.

B. LOTTERIES AND EQUAL DIVISION

Fix a round, a budget $B > 0$, and a pool $P \subseteq \{1, \dots, n\}$ of m researchers. The *equal division* over P grants each researcher in P the amount B/m . A *lottery* over P with $k \leq m$ winners selects k researchers from P uniformly at random and grants each winner B/k . Expected research output is evaluated as in Appendix A: given an allocation g , it is $\sum_i \varphi_i(g_i)$, and for a random allocation, the expectation is taken over the draw as well.

Proposition 2 (lotteries are outperformed by equal division). *For every pool P , budget B , and winner count $k \leq m$, the lottery's expected research output does not exceed the equal division's, with equality only if $k = m$. In particular, a lottery over the whole population, at any winner count, produces at most uniform funding's expected research output.*

Proof. Researchers outside P receive nothing under either scheme. For $i \in P$, the lottery's grant to i is a random variable $\tilde{g}_i$ equal to B/k with probability k/m and 0 otherwise, so $\mathbb{E}[\tilde{g}_i] = B/m$. By linearity, the lottery's expected research output is $\sum_{i \notin P} \varphi_i(0) + \sum_{i \in P} \mathbb{E}[\varphi_i(\tilde{g}_i)]$. Since φ_i is strictly concave (Lemma 1(ii)), Jensen's inequality gives $\mathbb{E}[\varphi_i(\tilde{g}_i)] \leq \varphi_i(\mathbb{E}[\tilde{g}_i]) = \varphi_i(B/m)$ for every $i \in P$, with equality only if $\tilde{g}_i$ is degenerate, that is, $k = m$. Summing over i yields the claim; taking $P = \{1, \dots, n\}$ yields the comparison with uniform funding. $\square$

Remark 1 (what the proposition does and does not cover). *The proposition compares expected research output within a fixed pool. It measures the effect of the randomization alone: whatever pool review selects, replacing the lottery within that pool by the equal division of the same money weakly increases expected research output. The proposition says nothing about the choice of pool, about the review*

ranking within the pool that the equal division also ignores, or about anything a lottery saves outside the model (review costs, proposal effort, bias). § 6 measures the first two and cites the literature on the third.

Remark 2 (minimum viable grant size). *The proposition needs divisible grants and diminishing returns at every grant size. Suppose instead that a project produces nothing unless its grant reaches a minimum m , and that $B < m|P|$ for a pool P . Equal division then gives every researcher in P less than m and produces no research output, while a lottery with $k = \lfloor B/m \rfloor$ winners, each receiving $B/k \geq m$, produces the research output of k funded projects. Where minimum project scale binds, the comparison of the proposition reverses, and this qualification applies wherever the main text compares lotteries with equal division.*

C. ROBUSTNESS TO THE PRODUCTION FUNCTION

This appendix asks which of our results depend on the harmonic-mean production function of §3. We embed that function in a family indexed by an exponent γ : expected research output is $A M_\gamma(K, R)$, where $M_\gamma(K, R) = ((K^\gamma + R^\gamma)/2)^{1/\gamma}$ is the power mean of capability and resources. At $\gamma = -1$ the power mean is the harmonic mean $2KR/(K + R)$, which is our production function up to the constant A . In the limit $\gamma \rightarrow 0$ (Cobb-Douglas) it is the geometric mean $\sqrt{KR}$, and a shortfall in either input can be offset by more of the other; as γ decreases the inputs grow harder to substitute; in the limit $\gamma \rightarrow -\infty$ (Leontief) it is $\min(K, R)$, and research output depends only on the scarcer input. We re-run the model's central comparisons across this family.¹⁸

The value of the review signal survives the whole family (Figure 12). At every production function tested, from Cobb-Douglas to Leontief, the signal's value increases as capability grows more heavy-tailed and as review grows more informative. The ordering of review's value by field (§5) is therefore not an artifact of our choice of production function. What changes with the production function is the scale. The more complementary the inputs, the more review is worth: under heavy-tailed capability with reliable review, the signal's value increases from about 16 percent of the research output that funding with review adds at Cobb-Douglas to about 47 percent at $\gamma = -3$ and about half at Leontief.¹⁹ Complementarity raises what targeting is worth, and with it the value of the information that targeting requires.

¹⁸ Settings: Appendix F.

¹⁹ 15.8, 47.1, and 51.4 percent. Cobb-Douglas lies below the two more complementary production functions at every tested tail and noise level at which the value is distinguishable from zero; $\gamma = -3$ and Leontief are within one standard error of each other at the noisiest review under heavy tails.

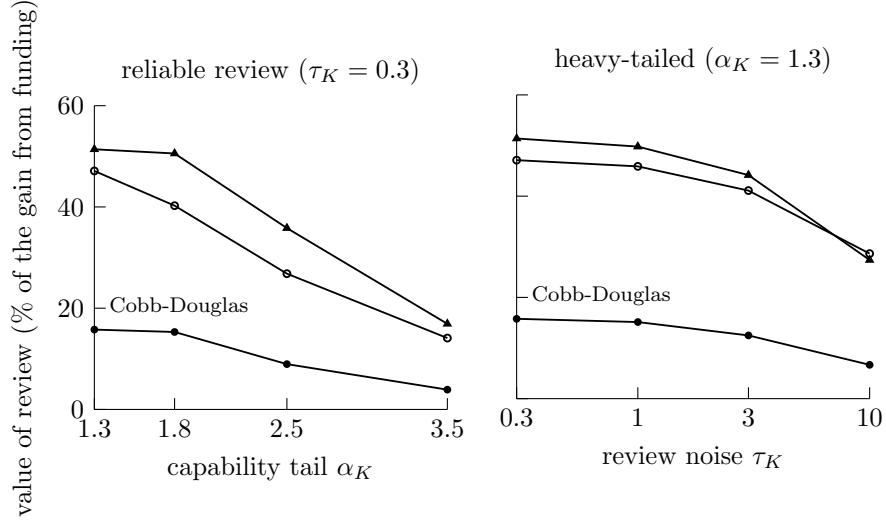


FIGURE 12. The structure of review’s value survives the production function. The value of the review signal, as a percent of the research output that funding with review adds over no funding, across the family of production functions: at every production function tested, the value increases as capability grows more heavy-tailed (left) and as review grows more informative (right), and its scale increases with complementarity. Filled circles: Cobb-Douglas; open circles: $\gamma = -3$; triangles: Leontief. Settings: Appendix F.

The timing results of Appendix E do not survive the substitutable end of the family (Table 3). Under Cobb-Douglas production the gain from planning ahead is indistinguishable from zero at every compounding rate, and the forward-looking funder’s timing stays within about a hundredth of even. At $\gamma = -3$, a production function more complementary than ours, both the gain from planning ahead and the shift of spending toward later rounds increase with the compounding rate and exceed their Cobb-Douglas values at every rate. The timing results hold where capability and resources are complements and vanish where they are substitutable.

How widely the informed funder spreads its grants across this family is reported in §6 (Figure 9).

D. SENSITIVITY OF THE VALUE OF TARGETING AND OF REVIEW

The main text varies the tail of the capability distribution, the budget, and review noise, and holds fixed the mean of the capability distribution, the association between capability and resources ($\rho = 0$), the tail of the resource distribution ($\alpha_R = 2$), the size of the applicant pool ($n = 50$), the noise of the resource signal ($\tau_R = 1$), and the form of the review signal (capability plus additive Gaussian noise, observed once). This appendix varies each of these and asks two questions:

TABLE 3. The gain from planning ahead and the forward-looking funder’s timing, by production function. Five rounds. The gain from planning ahead is the forward-looking funder’s gain over the myopic funder, both with review, as a percent of the research output that the forward-looking funder’s funding adds; the center of mass of spending is 0.5 when spending is even and larger when spending shifts toward later rounds. One hundred simulated populations per point; standard errors of the gain are at most 0.01 (Cobb-Douglas) and 0.4 ($\gamma = -3$) percentage points.

compounding rate ϵ	0.05	0.15	0.30	0.55	0.85
<i>Value of timing (percent of the gain from funding)</i>					
Cobb-Douglas ($\gamma = 0$)	-0.01	-0.01	0.00	0.01	0.03
$\gamma = -3$	0.05	0.56	1.96	4.56	6.52
<i>Center of mass of optimal spending</i>					
Cobb-Douglas ($\gamma = 0$)	0.499	0.500	0.501	0.505	0.512
$\gamma = -3$	0.511	0.540	0.577	0.623	0.660

whether the ordering of fields by the value of targeting and of review survives, and what single quantity, if any, organizes the results. Every run uses two rounds, the default budget $b = 1$, the myopic funders with and without review, the complete-information benchmark, and $M = 1,000$ particles; 50 simulated populations per setting, 100 in the noise sweeps. The value of targeting is the complete-information optimum’s gain over uniform funding as a share of uniform funding’s gain over no funding; the value of review is the share of the no-review shortfall that the review signal recovers, as in Figure 5.

Mean capability. A Pareto distribution with a fixed scale has mean $\alpha/(\alpha - 1)$, so at scale 1 the heavy-tailed field ($\alpha_K = 1.3$) has mean capability 4.3, the intermediate field ($\alpha_K = 2$) 2, and the evenly spread field ($\alpha_K = 3.5$) 1.4: a sweep of α_K at fixed scale varies mean capability along with inequality, and a higher mean makes resources more binding on its own. Table 4 compares that sweep with one in which the scale is set to $2(\alpha_K - 1)/\alpha_K$, so that every field has mean capability 2 (the realized means in the table fall short of 2 in the heavy-tailed field because the distribution has infinite variance; Appendix F). Holding the mean fixed narrows the heavy-tailed field’s advantage but preserves the ordering at every noise level; Figure 5 and every number in the main text use the fixed-mean sweep.

Review compared at equal informativeness. The same review noise τ_K is a more informative signal where capability is more dispersed, because the noise then moves fewer researchers past one another. Table 5 therefore compares fields at equal informativeness, measured by the rank correlation between the review score

TABLE 4. The value of review by field at a fixed Pareto scale and at a fixed mean capability. Share of the no-review shortfall recovered, at review noise $\tau_K = 0.3, 1$, and 3 ; 100 simulated populations per entry; $M = 1,000$ particles. The mean capability shown is the realized mean over the populations.

α_K	mean K	fixed scale 1			fixed mean 2			
		$\tau_K = 0.3$	1	3	mean K	$\tau_K = 0.3$	1	3
1.3	3.84	0.96	0.90	0.70	1.77	0.94	0.79	0.50
2	1.97	0.86	0.65	0.30	1.97	0.86	0.65	0.30
3.5	1.39	0.57	0.23	0.03	1.99	0.63	0.32	0.06

TABLE 5. The value of review by field at equal informativeness. Share of the no-review shortfall recovered when the rank correlation between the review score and capability is held at the stated value; fixed mean capability; interpolated within the noise sweep of Table 4.

α_K	rank correlation	0.8	0.55	0.3
1.3		0.94	0.80	0.50
2		0.85	0.65	0.32
3.5		0.66	0.46	0.18

and capability, interpolating within the fixed-mean sweep. At a rank correlation of 0.8, 0.55, and 0.3, the heavy-tailed field recovers 0.94, 0.80, and 0.50 of the shortfall, the intermediate field 0.85, 0.65, and 0.32, and the evenly spread field 0.66, 0.46, and 0.18. The ordering is unchanged; the gaps between fields are smaller than at equal noise.

The association between capability and resources, the resource tail, and pool size. Table 6 varies the three remaining population parameters one at a time, at the default review noise and at Pareto scale 1. The association between capability and resources is the one that matters. As the copula parameter ρ rises from -0.5 to 0.8 , the value of targeting falls in every field (from 0.96 to 0.69 in the heavy-tailed field, from 0.25 to 0.06 in the evenly spread one), and so does the value of review; the ordering of fields is unchanged at every ρ . The mechanism is the gap rule itself: when resources already sit with the capable, the gaps $cK_i - R_{i0}$ are small and even, and there is little for targeting, or for the information that targeting requires, to add. The tail of the resource distribution changes the value of targeting by at most a tenth and the value of review by less. Pool size raises the value of targeting in the heavy-tailed field, because a larger pool contains more of the rare researchers of very high capability, and raises the value of review in

TABLE 6. The value of targeting (upper block) and of review (lower block) by field as the copula parameter ρ coupling capability and resources, the tail of the resource distribution, and the size of the applicant pool vary. Review noise $\tau_K = 1$; Pareto scale 1; 50 simulated populations per entry; $M = 1,000$ particles. Default values: $\rho = 0$, $\alpha_R = 2$, $n = 50$.

α_K	copula parameter ρ				resource tail α_R			pool size n			
	-0.5	0	0.5	0.8	1.3	2	3.5	25	50	100	200
<i>Value of targeting (share of uniform funding's gain)</i>											
1.3	0.96	0.91	0.82	0.69	0.92	0.91	0.82	0.81	0.91	1.06	1.15
2	0.56	0.49	0.38	0.25	0.43	0.49	0.46	0.44	0.49	0.53	0.57
3.5	0.25	0.20	0.13	0.06	0.17	0.20	0.18	0.20	0.20	0.21	0.22
<i>Value of review (share of the no-review shortfall recovered)</i>											
1.3	0.92	0.90	0.88	0.85	0.88	0.90	0.92	0.87	0.90	0.94	0.95
2	0.67	0.65	0.62	0.56	0.65	0.65	0.65	0.60	0.65	0.71	0.74
3.5	0.24	0.22	0.16	0.01	0.24	0.22	0.21	0.14	0.22	0.27	0.32

the evenly spread field, where a larger pool gives review more to separate; neither effect changes the ordering.

The resource signal. The main text holds the resource signal at noise $\tau_R = 1$. Table 7 varies it from nearly exact ($\tau_R = 0.1$) to nearly uninformative ($\tau_R = 10$) and removes it altogether. Resource information matters least where review matters most. In the heavy-tailed field the no-review shortfall moves from 0.44 to 0.48 of the complete-information gain across the whole range, and the value of review from 0.79 to 0.76: the researchers who matter there are identified by capability, and resources add little to their identification. In the evenly spread field a nearly exact resource signal cuts the no-review shortfall from 0.15 to 0.09, because once resources are known, research output identifies capability; there the resource signal does a part of what review does, and the value of review falls from 0.39 with an exact resource signal to 0.24 with none. The ordering of fields is unchanged at every τ_R .

Multiplicative review noise. The main text's review noise is additive, so a given τ_K is a smaller relative error for a researcher of high capability than for one of low capability, and the question is whether that form favors the heavy-tailed field. Table 8 replaces the score by $K_i \exp(e_i)$ with e_i Gaussian of standard deviation s , a constant relative error, with the funder knowing the form. The form does matter, in the direction the question anticipates. At equal informativeness, multiplicative noise costs the heavy-tailed field most: at a rank correlation of 0.65 between score and capability, review recovers 0.66 of the shortfall under multiplicative noise against 0.87 under additive noise in the heavy-tailed field,

TABLE 7. The resource signal. The no-review shortfall (upper block) and the value of review (lower block) as the noise of the resource signal τ_R varies from nearly exact to nearly uninformative, and with no resource signal; review noise $\tau_K = 1$; fixed mean capability; 50 simulated populations per entry; $M = 1,000$ particles.

α_K	resource-signal noise τ_R					none
	0.1	0.3	1	3	10	
<i>No-review shortfall (share of the complete-information gain)</i>						
1.3	0.45	0.44	0.45	0.47	0.48	0.48
2	0.25	0.25	0.27	0.29	0.30	0.30
3.5	0.09	0.09	0.11	0.14	0.15	0.15
<i>Value of review (share of the no-review shortfall recovered)</i>						
1.3	0.79	0.79	0.79	0.77	0.77	0.76
2	0.68	0.67	0.65	0.61	0.59	0.59
3.5	0.39	0.37	0.31	0.26	0.24	0.24

TABLE 8. Multiplicative review noise. The review score is $K_i \exp(e_i)$ with e_i Gaussian of standard deviation s , and the funder knows the form; fixed mean capability; 50 simulated populations per entry; $M = 1,000$ particles. The upper block gives the informativeness of the score at each s .

α_K	noise s				
	0.1	0.3	0.6	1	1.5
<i>Rank correlation between score and capability</i>					
1.3	0.96	0.83	0.65	0.48	0.35
2	0.93	0.73	0.51	0.34	0.24
3.5	0.85	0.55	0.33	0.21	0.14
<i>Value of review (share of the no-review shortfall recovered)</i>					
1.3	0.96	0.88	0.67	0.43	0.27
2	0.89	0.71	0.42	0.21	0.09
3.5	0.68	0.39	0.15	0.05	0.02

0.61 against 0.75 in the intermediate field, and 0.49 against 0.55 in the evenly spread field, because a constant relative error is a large absolute error for exactly the researchers who matter most in a heavy-tailed field. The ordering of fields holds at every s and at every matched informativeness, and so does the main text’s statement that noisy review retains most of its value under heavy tails: at $s = 0.6$, review retains 69 percent of its maximum value in the heavy-tailed field, 47 percent in the intermediate field, and 22 percent in the evenly spread field. The heavy-tailed field’s advantage is smaller under multiplicative noise than the main text’s additive figures show, and it does not disappear.

TABLE 9. Review observed once against review redrawn every round. Share of the no-review shortfall recovered over two rounds at review noise $\tau_K = 0.3, 1$, and 3 ; fixed mean capability; 100 (once) and 50 (redrawn) simulated populations per entry; $M = 1,000$ particles.

α_K	observed once (main text)			redrawn every round		
	$\tau_K = 0.3$	1	3	0.3	1	3
1.3	0.94	0.79	0.50	0.95	0.82	0.52
2	0.86	0.65	0.30	0.88	0.68	0.32
3.5	0.63	0.32	0.06	0.66	0.36	0.07

Review observed once against review redrawn every round. In the main text each researcher’s review score is observed once and reused in every round, so a reviewer’s misjudgment persists (§3). Table 9 redraws the score every round with independent noise, so that repeated review accumulates information the way the record does. Over two rounds the second review adds 0.01 to 0.04 to the share of the shortfall recovered, most where review is noisiest, and the ordering of fields is unchanged. Over twenty rounds the difference is larger: in the intermediate field the review funder’s shortfall falls from 8 percent of the complete-information gain in round one to 3 percent by round twenty when the score is observed once (Figure 4), and to 0.8 percent when it is redrawn every round; in the heavy-tailed field, to 2 percent and to 0.7 percent. The no-review funder’s shortfall in the same runs falls from 46 to 19 percent (default) and from 60 to 14 percent (heavy-tailed) either way. Repeated independent review would therefore strengthen the main text’s comparison, and the once-observed score is the conservative case.

Review that partly reflects resources. The main text’s review signal observes capability with noise. Real review may also reflect the resources a researcher visibly has. Table 10 replaces the signal by $K_i + \beta R_{i0} + \xi_i$, with the funder knowing β and updating accordingly. Contamination costs least where capability is heavy-tailed: at $\beta = 1$, where the score weights resources as heavily as capability, the heavy-tailed field still recovers 0.70 to 0.76 of the shortfall, against 0.79 to 0.94 for a clean signal. It costs most where capability is evenly spread, where at $\beta = 1$ review recovers 0.17 to 0.26 of the shortfall, against 0.32 to 0.63 for a clean signal. The reason is the one behind Figure 5: the researchers whom review must find in a heavy-tailed field stand so far apart in capability that a resource term does not hide them.

The inequality of the gaps organizes the results. The runs above vary the population parameters in five ways. One quantity summarizes what they do to the value of targeting: how unequally the optimal grants, the gaps $\max(cK_i - R_{i0}, 0)$ at

TABLE 10. The value of review when the review score partly reflects resources. Share of the no-review shortfall recovered when the score is $K_i + \beta R_{i0}$ plus noise, for $\beta = 0$ (the main text’s signal), 0.25, 0.5, and 1; fixed mean capability; 50 simulated populations per entry (100 at $\beta = 0$); $M = 1,000$ particles.

α_K	$\tau_K = 0.3$				$\tau_K = 1$			
	$\beta = 0$	0.25	0.5	1	$\beta = 0$	0.25	0.5	1
1.3	0.94	0.92	0.88	0.76	0.79	0.77	0.75	0.70
2	0.86	0.82	0.75	0.55	0.65	0.62	0.59	0.52
3.5	0.63	0.55	0.42	0.26	0.32	0.27	0.23	0.17

the population’s budget, are distributed across the field. Across all 159 population settings in this appendix (mean capability, association, resource tail, pool size, and the two noise sweeps), the logarithm of the value of targeting is a nearly linear function of the Gini coefficient of the optimal grants, with $R^2 = 0.96$ (Figure 13); adding the ratio of mean capability to mean resources raises R^2 to 0.99. Settings that differ in the capability tail, the association between capability and resources, the resource tail, and pool size, but agree on the inequality of the gaps, have the same value of targeting. The value of review depends on the same quantity together with the signal’s informativeness: the log-odds of the share recovered, regressed on the Gini coefficient of the optimal grants and the log-odds of the rank correlation between score and capability, has $R^2 = 0.90$, and adding the capability tail as a separate variable raises it by less than 0.01. Capability inequality is the main driver of gap inequality in every parameter sweep we considered, which is why the main text can speak of it directly; the association between capability and resources is the second driver, and it works through the same channel. Two cautions apply to using this quantity. The gaps depend on the budget, through c , as well as on the population, so the inequality of the gaps is a property of a field and a funder together; and it is an unobserved quantity, computed from capabilities and resources the funder does not observe, so it organizes the model rather than serving as a statistic a funder can estimate directly. The main text’s Discussion lists the observable quantities that stand in for it.

E. THE TIMING OF FUNDING

The funder holds a fixed budget for the whole horizon, and between rounds two things change: researchers produce research output, which the funder observes, and capabilities compound. In the main text the funder spends its budget in equal parts each round; this appendix asks whether it should spend early, evenly, or late, and how much the choice matters.

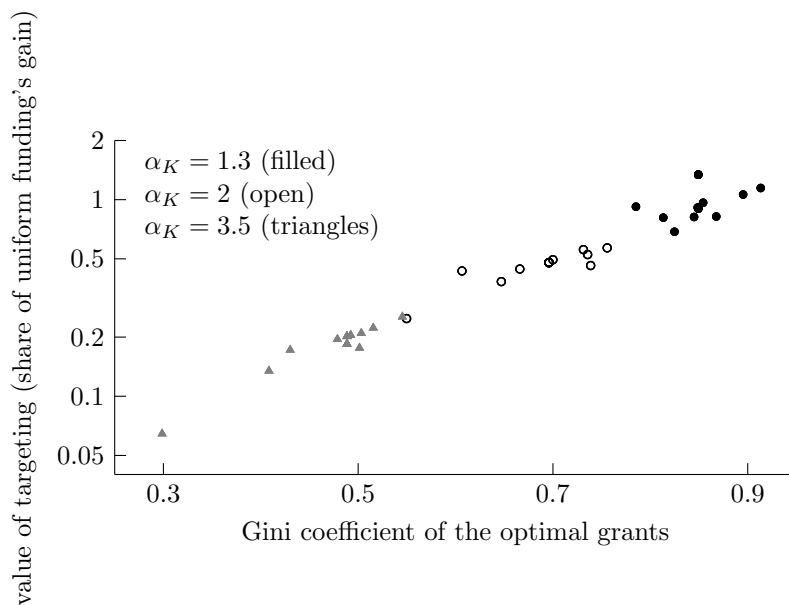


FIGURE 13. The inequality of the gaps organizes the value of targeting. Each point is one setting of Appendix D: the value of targeting, on a logarithmic scale, against the Gini coefficient of the optimal grants at that setting's budget. The 159 settings vary the capability tail (marker), the correlation between capability and resources, the tail of the resource distribution, the size of the applicant pool, and the scale of the capability distribution; the logarithm of the value of targeting is linear in the Gini coefficient with $R^2 = 0.96$.

The strongest argument for spending early concerns fields whose researchers have few resources. A researcher without resources produces nothing, so the funder observes nothing and no capability grows; money held for later rounds therefore produces neither evidence nor growth in the meantime. On this argument a funder facing such a field should spend at once. The argument fails. A funder that spends everything in the first round allocates before observing any research output, and knows exactly as little as a funder that spends everything in the last round; in the extreme case of a field with no resources at all, the two produce identical expected research output. What the argument does establish is a complementarity: early spending enables observation and growth, and late spending exploits what early spending revealed. This complementarity favors putting some money early. It does not favor putting most of it early.²⁰

²⁰ Among timings fixed in advance, the best timing that spends more than an even share in the early rounds loses to even spending by 1.6 percent of even spending's research output at negligible compounding and by 3.5 percent at the default compounding rate (six rounds, community with few resources).

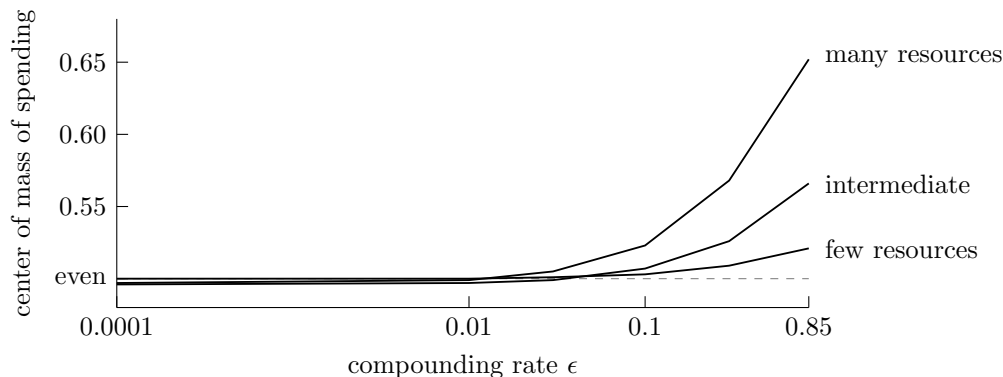


FIGURE 14. The compounding rate, not the scarcity of baseline resources, sets how far spending shifts toward later rounds. The center of mass of spending (0.5 is even; higher is later) as the compounding rate ϵ increases, for three baseline resource scales. Settings: Appendix F.

Our simulations bear this out: under complementarity, observation comes first and money second. Wherever capabilities compound at more than a negligible rate, the forward-looking funder spends less than an even share in the first round and spends the bulk once research output has revealed who is capable; spending late pays because money spent later is spent with more information, and because early research output compounds on its own (Figure 14). Scarce baseline resources raise the early share and decreases the intensity of the effect, but does not reverse it.²¹

How much a funder can learn from its own grants depends on their size. A researcher on a small grant produces research output nearly proportional to the grant and nearly independent of capability, so a funder whose grants are small learns almost nothing it can exploit later: in a field whose researchers have few resources, with no review signal, such a funder gains almost nothing over uniform funding. Large grants change this. As grants approach the scale of capability itself, research output separates the capable from the rest, and the funder recovers most of the value of discrimination, with timing that spends a small share early, so that research output can be observed, and the rest once the funder has learned who is capable (Figure 15).²²

²¹ Every parameter setting in which the forward-looking funder spends more than an even share early has a compounding rate $\epsilon \leq 0.03$, and there the center of mass of spending is within 0.005 of even.

²² In a community whose researchers have few resources, with heavy-tailed capability and no review signal, over six rounds, the Bayesian funder's gain over uniform funding increases from a third of one percent of uniform funding's research output at the default grant size to nine percent at six times that size and ten percent at twelve times, roughly the gain a reliable review signal delivers at the default size.

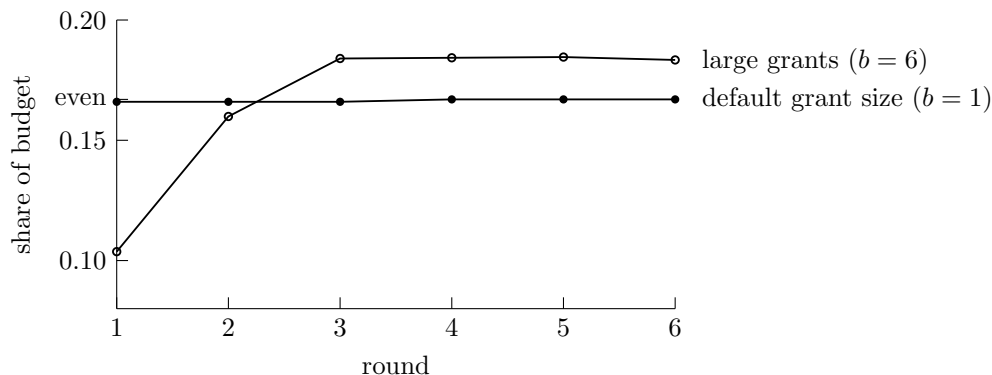


FIGURE 15. With large grants the funder spends a small share first and the rest once it has observed research output. The share of the budget spent in each of six rounds, in a field whose researchers have few resources, with heavy-tailed capability and no review signal, at the default grant size (filled) and at six times that size (open). Settings: Appendix F.

The shift of spending toward later rounds comes from the research output researchers produce whether or not they are funded: that research output compounds and reveals capability while the funder waits. Growth from grant-added research output alone would shift spending slightly toward earlier rounds, since earlier grants have longer to compound.²³

Finally, planning ahead is worth less than the signal provided by grant peer review, in every setting we test. Across all horizons and compounding rates in our sweeps, the forward-looking funder's gain over the myopic funder, which spends equal installments each round, stays below the review signal's value: at the default compounding rate and below it is smaller by an order of magnitude or more, and it approaches the signal's value only at the joint extreme of the highest compounding rate and the longest horizon we consider (Figure 16).²⁴ The gain increases roughly with the square of the horizon over short horizons and levels off beyond five rounds. And it exists only where capability and resources are complements, as in our production function: under Cobb-Douglas production the gain from planning ahead is indistinguishable from zero and the forward-looking funder's timing stays within about a hundredth of even (Appendix C, Table 3).

²³ With growth from unfunded research output switched off, the center of mass of spending is 0.48 to 0.49, below even; with growth from grant-added research output switched off, 0.53 to 0.68 (five rounds).

²⁴ As a share of the research output that the forward-looking funder's funding adds, the gain from planning ahead ranges from indistinguishable from zero to 4.5 percent across the thirty tested parameter settings; the review signal's value in the same settings ranges from 7 to 27 percent.

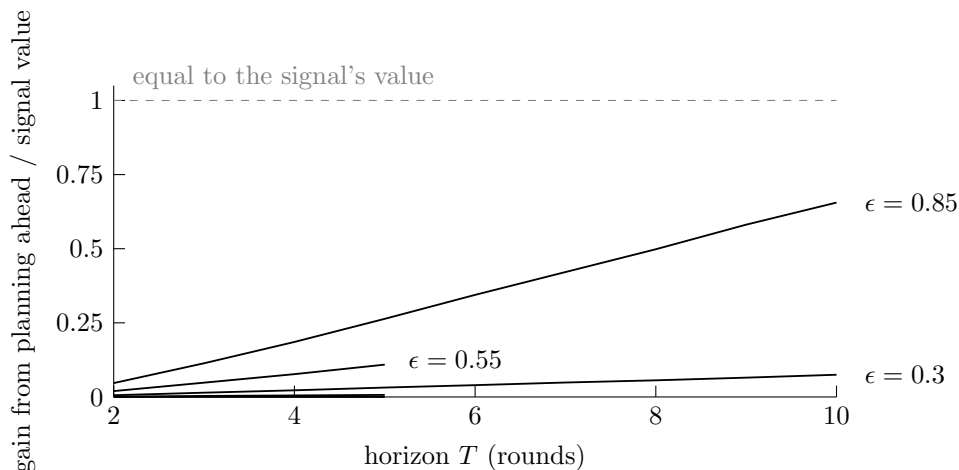


FIGURE 16. Choosing whom outweighs choosing when. The gain from planning ahead, as a share of the review signal’s value, by horizon and compounding rate ϵ . In every setting the gain is below the signal’s value, and it approaches that value only at the highest rate and the longest horizon. Settings: Appendix F.

For a funder thinking about how to maximize their impact, choosing whom to fund outweighs choosing when.

Three limits bound these results, and they are the reason the timing analysis sits in an appendix rather than the main text. First, the forward-looking funder is a certainty-equivalent planner (§3): it treats future research output as equal to its expectation and does not value the information its own grants will generate, so it is not the Bayes-optimal sequential policy, and its gain is a lower bound on what planning ahead could be worth. Second, the gain combines two changes at once, when the money is spent and whom it reaches, because a funder that anticipates capability growth also values recipients differently; we do not separate the two, and “gain from planning ahead” names their sum. Third, the objective counts research output within the horizon only and assigns no value to the capability that remains at its end. A terminal value for capability would credit the capability that late grants build, which the horizon objective ignores, and also the compounded capability that early grants leave behind; we have not measured which effect is larger, so the size of the shift toward later rounds should be read as specific to the horizon objective.

F. SIMULATION SPECIFICATION

Budget and productivity. Over the whole horizon, the funder’s total budget is $bn\mathbb{E}[R_0]$, where $\mathbb{E}[R_0]$ is the expected baseline resources of one researcher, so at $b = 1$ the budget equals the community’s expected baseline resources for

one round. The budget does not depend on the horizon T . Strategies that do not plan ahead spend $1/T$ of the budget each round. For the timing runs of Appendix E (Figures 15 and 14) the budget is scaled against capability instead, $b n \mathbb{E}[K_0]$, so that a community with few resources does not zero the budget. The simulations set the productivity constant $A = 1$. The code parameterizes the budget as $2 b' n \mathbb{E}[R_0]$; its b' is half of the b reported here.

Default parameters. Unless a figure’s settings below say otherwise: $T = 2$ rounds; $n = 50$ researchers; capability and baseline resources drawn from Pareto distributions with scale 1 and tail parameters $\alpha_K = \alpha_R = 2$; copula parameter $\rho = 0$; compounding rate $\epsilon = 0.1$; budget scale $b = 1$; review noise $\tau_K = 1$ and resource-signal noise $\tau_R = 1$; seed-grant fraction 0.25 where seed grants are used.

Population draws, particles, and heavy tails. Capability and resources are coupled through a Gaussian copula: standard normal draws with correlation ρ are mapped through the normal distribution function to uniforms and then through the Pareto quantile functions. A Pearson correlation between two Pareto variables is undefined when either tail parameter is at most 2, so ρ is a copula parameter and not a correlation of the draws; the rank correlation it induces is $(6/\pi) \arcsin(\rho/2)$, which is 0.48 at $\rho = 0.5$ and 0.79 at $\rho = 0.8$. Below a tail parameter of 2 the Pareto distribution has infinite variance and sample means converge slowly: at $\alpha_K = 1.3$ with the scale set for a mean of 2, the mean capability across 100 populations of 50 is 1.77 and the median population has mean 1.48, because the rare draws that carry the mean are absent from most samples. Heavy-tailed results are therefore reported with standard errors across populations, and Table 11 reports medians beside means.

Bayesian funders compute each researcher’s posterior by importance sampling: M particles (K, R_0) are drawn from the prior afresh in each round and weighted by the full likelihood of the researcher’s record and signals to date. With few particles the posterior can rest on a handful of them, and rare regions of the prior, where the most capable researchers of a heavy-tailed field sit, can be missed. Table 11 checks this. Raising M from 200 to 1,000 raises the value of review by 0.01 to 0.02 in every field at every noise level (the posteriors at $M = 200$ understate the capability of the most capable researchers), and raising it further to 5,000 changes the value of review by at most 0.005, within one standard error everywhere. The value of targeting uses no particles. Figures 5 and 4, the sensitivity analysis of Appendix D, and Figure 10 use $M = 1,000$; the remaining simulation figures use $M = 200$, and by the table their values of review are understated by about 0.01 to 0.02 of the no-review shortfall, which does not change any comparison they support.

TABLE 11. Convergence in the number of particles. The value of review (share of the no-review shortfall recovered; mean over 100 populations, with the median in parentheses) and the effective sample size of the round-one posteriors (median across researchers and populations, with the tenth percentile in parentheses), by field, review noise, and number of particles M . Two rounds, $b = 1$, mean capability 2; the same 100 populations at every M . The value of targeting uses no particles and is unchanged.

α_K	τ_K	$M = 200$	$M = 1000$	$M = 5000$
<i>Value of review, mean (median)</i>				
1.3	0.3	0.91 (0.94)	0.94 (0.96)	0.94 (0.97)
1.3	1	0.77 (0.83)	0.79 (0.84)	0.79 (0.85)
1.3	3	0.48 (0.56)	0.50 (0.58)	0.50 (0.53)
2	0.3	0.84 (0.87)	0.86 (0.89)	0.87 (0.89)
2	1	0.63 (0.67)	0.65 (0.69)	0.65 (0.69)
2	3	0.28 (0.26)	0.30 (0.26)	0.29 (0.27)
3.5	0.3	0.62 (0.65)	0.63 (0.66)	0.64 (0.67)
3.5	1	0.31 (0.32)	0.32 (0.33)	0.32 (0.33)
3.5	3	0.05 (0.04)	0.06 (0.05)	0.06 (0.04)
<i>Effective sample size, median (tenth percentile)</i>				
1.3	0.3	81 (5)	396 (28)	1998 (136)
1.3	1	122 (15)	618 (73)	3081 (369)
1.3	3	142 (36)	711 (164)	3559 (796)
2	0.3	74 (6)	368 (33)	1843 (166)
2	1	124 (18)	616 (90)	3079 (447)
2	3	146 (46)	732 (227)	3667 (1137)
3.5	0.3	92 (12)	460 (56)	2300 (294)
3.5	1	136 (39)	685 (193)	3417 (955)
3.5	3	153 (67)	771 (314)	3857 (1558)

Scoring and comparison. Each strategy is scored by the sum of expected research output along its run, which equals realized research output averaged over repetitions of the same allocations; this removes simulation noise from comparisons without favoring any strategy. Every configuration is run on independently drawn populations, with the same random draws (populations, signals, and realized research output) used for every strategy, so that comparisons between strategies are paired. Results are reported as shares of a named comparison, stated in each figure: most often the research output that funding adds (a strategy’s expected research output minus the expected research output under no funding), and otherwise the no-review shortfall or the optimal allocation’s gain. Standard errors, where shown, are across simulated populations.

Settings by figure.

Placing programs on the budget axis. A round of the model corresponds to a year of a program’s awards, and the regime map’s budget b is the total over its two rounds, so a program whose annual awards to a field are a fraction r of the field’s annual research spending from every other source, which stands in for the baseline resources of the field’s researchers, sits at $b = 2r$. The estimate counts money and not time, and treats each funder’s field as the disciplines it mainly funds, so it is an order of magnitude; the bands in the figure span a factor of about two around each placement. United States universities spent \$108.8 billion on research and development in fiscal year 2023, \$55.2 billion of it in the life sciences, and received \$33.1 billion from the Department of Health and Human Services, chiefly the National Institutes of Health, and \$6.7 billion from the National Science Foundation (National Center for Science and Engineering Statistics 2025): a biomedical agency of that size has r near 1.5 (\$33 billion against \$22 billion from other sources), so b near 3, at the top of the map; a science agency of that size funding the other half of academic research has r near 0.15 and b near 0.3. The European Research Council’s budget of 16 billion euros over 2021 to 2027 (European Research Council 2025), about 2.3 billion a year, against 86 billion euros of higher-education research spending in the European Union in 2024 (Eurostat 2025), gives r near 0.03 and b near 0.05. A private foundation awarding on the order of \$100 million a year across several fields has r at or below 0.01 in any one of them, and b at or below 0.02.

Code. The model, the sweeps, and the scripts that produce every figure and number in the paper are in the project repository; the archived version will be cited at publication.

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Figure	Settings that differ from the defaults
5	Review noise τ_K from 0.05 to 20; $\alpha_K \in \{1.3, 2, 3.5\}$ with the Pareto scale set to $2(\alpha_K - 1)/\alpha_K$ so that mean capability is 2 in every field; 100 populations per point; $M = 1,000$. The no-review shortfall relative to complete information is 44, 26, and 11 percent of the research output that complete-information funding adds, at the three tail parameters.
4	$T = 20$; 50 populations; $M = 1,000$. No-review is the myopic funder without review; with review is the myopic funder with a single review score observed at the start. The shortfall is the funder's expected research output below the complete-information optimum in that round, as a share of the optimum's gain over no funding. Under heavy-tailed capability ($\alpha_K = 1.3$, mean 2) the no-review shortfall falls from 60 to 14 percent against 6 to 2 percent with review, and the grant correlations run 0.12 to 0.55 against 0.89 in round one.
6	True noise $\tau_K = 3$; believed noise from 0.1 to 10; $\alpha_K \in \{1.3, 2\}$; 200 populations. Each negative value is within two standard errors of zero.
15	$T = 6$; community with few resources (Pareto scale for resources 0.001); $\alpha_K = 1.3$; no review signal ($\tau_K = 100$); $\epsilon = 10^{-4}$; budget scaled against capability; $b = 1$ (50 populations) and $b = 6$ (24 populations, standard error of each share at most 0.006).
14	$T = 5$; forward-looking funder with review; budget scaled against capability; resource scales 0.001, 0.3, and 3; ϵ from 10^{-4} to 0.85; 200 populations. Five resource scales were run; the figure shows three, and the other two lie between them.
16	T from 2 to 10 (to 5 for $\epsilon \in \{0.05, 0.15, 0.55\}$); $\epsilon \in \{0.05, 0.15, 0.3, 0.55, 0.85\}$; 64 to 200 populations per point. The gain is the forward-looking funder's research output over the myopic funder's, both with review; negative gains, all within noise of zero, are shown as zero.
7	Review-informed forward-looking funder; seed-grant fraction from 0 to 0.75; heavy-tailed field ($\alpha_K = 1.3$) with reliable review ($\tau_K = 0.3$) at $b \in \{0.2, 1, 2\}$, and the intermediate field at $b = 1$; 200 populations. The research output lost is reported as a share of the review-informed funder's gain over no funding in the same setting.
8	Single round; $n = 50$; a fifth of the population funded; review score K_i plus Gaussian noise of standard deviation τ ; expected research output computed exactly given each allocation; 2,000 populations per point (Monte Carlo standard error at most 0.002). Left: $\alpha_K = 1.3$, $b = 0.2$. Right: $\alpha_K = 3.5$, $b = 1$.
12, Table 3	Production functions with $\gamma \in \{0, -3\}$ and Leontief; 100 populations per point (50 for Leontief); $T = 2$ for the signal value and $T = 5$ for the timing table. These runs use the greedy allocator, the only one implemented for the general family (the harmonic-mean runs use exact water-filling), so their magnitudes are compared with one another and not with the body's figures. Leontief allocations use a finer allocation step, with ties at the kink broken by fill order.
9	Single round; review-informed funder; γ from 0 to -12 and Leontief; three field-and-budget combinations ($\alpha_K = 1.3$, $b = 0.2$; $\alpha_K = 2$, $b = 2$; $\alpha_K = 3.5$, $b = 2$); 200 populations per point, 400 at the refinement points of the evenly spread curve.
13, Tables 4 to 10	$T = 2$; $b = 1$; myopic funders with and without review and the complete-information benchmark; $\rho \in \{-0.5, 0, 0.5, 0.8\}$; $\alpha_R \in \{1.3, 2, 3.5\}$; $n \in \{25, 50, 100, 200\}$; Pareto scale 1 or set to give mean capability 2; review score $K_i + \beta R_{i0}$ plus noise with $\beta \in \{0, 0.25, 0.5, 1\}$; resource-signal noise $\tau_R \in \{0.1, 0.3, 1, 3, 10\}$ and no resource-signal multiplicative review noise for $\beta = 0$ and K_{i0} with